Slodowy Slices and Their Quantization

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Overview

Nilpotent orbits of Lie algebras are important objects to study in representation theory, algebraic geometry, and other related fields. Among them, the Slodowy slice of a nilpotent orbit not only provides numerous examples of algebraic varieties (such as ADE singularities), but also can be served as a source of many important structures (such as W-algebras, Yangians, etc.). In this paper, we study Slodowy slices and their quantization in detail. First, we give the definition of Slodowy slices and define their Poisson structure in two different ways, namely by restricting the given Poisson structure to symplectic leaves and via Hamiltonian reduction. Then we realize the Poisson structure of Slodowy slices by BRST cohomology and quantize them by means of quantized BRST cohomology, thereby obtaining W-algebras. Finally, we present another definition of W-algebras via the Whittaker model and show that it is isomorphic to the previous definition.

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Chapter 1

Introduction

Slodowy slices are classical objects in geometric representation theory, and they play an important role in the study of nilpotent orbits. In type A, there are many important results on Slodowy slices. For example, Maffei showed in [Maf05] that a resolution of a certain closed subvariety in a Slodowy slice, called a Slodowy variety, is isomorphic to a Nakajima quiver variety $\mathfrak{m}(d, v)$. Moreover, this closed subvariety in the Slodowy slice is isomorphic to the Lagrangian subvariety $\mathfrak{m}^1(d, v)$ of $\mathfrak{m}(d, v)$. The following diagram is commutative:

$$\begin{tikzcd} \mathfrak{m}(d,v) \arrow[r, "\widetilde{\varphi}"] \arrow[d, "\pi"'] & \widetilde{\mathcal{S}}_{d,x} \arrow[d, "\mu_d"'] \\ \mathfrak{m}^1(d,v) \arrow[r, "\varphi_1"'] & \mathcal{S}_{d,x}. \end{tikzcd}$$

Using Maffei's result, Losev proved in [Los12] that the quantum Hamiltonian reduction associated with a type A quiver variety is isomorphic to the quantization of a type A parabolic Slodowy slice, namely a type A parabolic W-algebra. However, Maffei's isomorphism is described by a recursive characterization rather than by an explicit formula. Similarly, Losev's proof does not write down the isomorphism explicitly.

To overcome these limitations, Mirković and Vybornov developed a more ex-

plicit geometric method in [MV03]. They constructed type A Nakajima quiver varieties by affine Grassmannian slices. In this way, they showed that in type A, the singularities of quiver varieties, nilpotent orbits, and affine Grassmannians are equivalent. Moreover, in [MV08], they related quiver varieties, conjugacy classes of matrices, and the Beilinson–Drinfeld Grassmannian. This gives a compactification of quiver varieties and decomposes the affine Grassmannian into a disjoint union of quiver varieties. Their construction is close in spirit to Maffei's work, but it is more explicit. In this sense, it can be viewed as a classical analogue of the Brundan–Kleshchev presentation; see [BK06]. It is also closely related to a partly conjectural viewpoint on symplectic dual pairs; see [Bra+10, Remark 1.5].

From this point of view, there are still many open questions in this area. For example, one may ask whether similar results hold beyond type A, how to study the representations of the W-algebras obtained after quantization, and how the geometric properties of quiver varieties, Slodowy slices, and affine Grassmannians correspond to each other in concrete terms. Therefore, the study of Slodowy slices is especially important.

In § 3.1, following Slodowy's monograph [Slo80], we define Slodowy slices. For a nilpotent element $e \in N$, the Jacobson–Morozov theorem gives an $\mathfrak{sl}_2$-triple (e, h, f). We then prove that the affine space

$$S = e + \operatorname{Ker}(\operatorname{ad} y)$$

is a special transversal slice to the adjoint orbit through e. We call this slice a Slodowy slice. In § 3.3, we give the Poisson structure on a Slodowy slice. By a classical result of Weinstein [Wei83], this Poisson structure can be obtained by restriction. Gan and Ginzburg gave a richer construction in [GG02]: they realized Slodowy slices as Poisson varieties by Hamiltonian reduction. Along this direction, a main result of Lehn et al. [LNS12] gives a necessary and sufficient condition under which a Slodowy slice is identified with the universal Poisson

deformation of its nilpotent part.

To study the quantization of the Poisson structure on a Slodowy slice, in § 4.1 we prove that a certain cohomological Poisson algebra is isomorphic to the Poisson structure on the Slodowy slice. This gives the BRST realization of the Slodowy slice. Then, in § 4.2, we give the quantum version of BRST cohomology. This gives a quantization of the Slodowy slice. Finally, in § 4.3, we explain that this quantization is the finite W-algebra defined by the Whittaker model; see [Ara07b]. The quantization of Slodowy slices has led to many results. For a conic symplectic singularity X, Namikawa proved in [Nam06; Nam10] that the Poisson deformation functor is representable. Losev showed that the universal Poisson deformation can be quantized, and that this quantization has a useful universal property; see [Los12]. More recently, Ambrosio and Topley introduced in [Amb+23] a quantization functor for flat Poisson deformations, and they explained that this universal property is equivalent to the representability of the quantization functor. At the same time, in [AT23], they lifted this picture to the equivariant setting and completely described when a finite W-algebra is the universal filtered quantization of the slice. This gives an almost complete classification of filtered quantizations of nilpotent Slodowy slices.

The generators and relations of W-algebras have long been a central topic. In the finite-dimensional classical W-algebra case, this can be studied by Yangians. Brundan and Kleshchev used shifted Yangians to give a presentation of the finite W-algebras associated with nilpotent matrices in the general linear Lie algebra; see [BK06]. Motivated by this, Kang Lu et al. constructed in [LWW25] a family of truncated shifted twisted Yangians of type AI, denoted by $y^{+}_{n,\ell}(\sigma)$. This family of algebras is defined by generators and relations, and it generalizes the work of Brundan and Kleshchev in a very different way. They further applied the theory of universal equivariant quantization of conic symplectic singularities and proved

that $y^+_{n,\ell}(\sigma)$ is isomorphic to a finite W-algebra. Thus it quantizes the corresponding BCD-type Slodowy slices.

There is also a parallel quantization theory for another classical geometric object, namely affine Grassmannian slices. Let G be a complex semisimple group, and let $\mathrm{Gr} = G((t^{-1}))/G[t]$ be the thick affine Grassmannian. For dominant coweights $\lambda \geq \mu$, the transversal slice $\mathrm{Gr}^{\bar{\lambda}}_{\mu}$ is an affine Poisson variety. Kamnitzer et al. gave in [Kam+14] an explicit quantization of its coordinate ring by a quotient of a shifted Yangian. On the other hand, Mirković and Vybornov proved in [MV08] that there is an isomorphism between Slodowy slices in $\mathfrak{sl}_m$ and slices in the affine Grassmannian of GL_n. Therefore, quotients of shifted Yangians for $\mathfrak{gl}_n$ can be viewed as quantizations of affine Grassmannian slices through their relation with W-algebras. In [WWY20], Webster et al. established a quantum version of the Mirković–Vybornov isomorphism. It relates the intersection of a Slodowy slice in $\mathfrak{gl}_N$ with a nilpotent orbit closure to a slice in a spherical Schubert variety in the affine Grassmannian of PGL_n. The former is quantized by a parabolic W-algebra, and the latter is quantized by a truncated shifted Yangian. In summary, these results reveal rich triangular relations among Nakajima quiver varieties, Slodowy slices, and affine Grassmannian slices, both at the classical level and at the quantum level.

Beyond type A, Henderson and Licata showed in [HL14] that, under a diagram automorphism, the fixed-point subvariety of a quiver variety is a disjoint union of quiver varieties of the split-quotient quiver. As an application, they obtained isomorphisms between two-row Slodowy slices of types C and D and the corresponding type D quiver varieties. These isomorphisms are compatible with the natural reductive group actions. In each case, they also constructed an isomorphism between the slice itself and an affine variety, and an isomorphism between the Springer fibre over the base point of the slice and a projective variety. These

results generalize Maffei's results in type A.

The many results and open questions related to Slodowy slices motivate the present study. In this paper, we follow the lines of [Slo80; GG02; AM16]. First, we give the definition of a Slodowy slice, and we define its Poisson structure in two different ways: by restricting the given Poisson structure to symplectic leaves and by Hamiltonian reduction. Second, we realize the Poisson structure of a Slodowy slice by BRST cohomology, and then quantize it by quantum BRST cohomology. This construction gives a finite W-algebra. Finally, we state another definition of W-algebras by the Whittaker model, and we explain that it is isomorphic to the preceding definition. Since Slodowy slices corresponding to two-row split nilpotent orbits are widely studied, we also give an example in $\mathfrak{sl}_4$.

Chapter 2

Preliminaries

In this chapter, we recall some necessary background from geometry and algebra. More precisely, in 2.1 we recall the definition of affine tangent spaces and the differential of related morphisms, which will be essential for defining transversal slices. In 2.2, we recall the definition of semisimple groups and the definitions of exp and log, which will be used directly in the actions of algebraic groups on various objects. In 2.3, we recall the construction of canonical symplectic structures in symplectic geometry and the definitions related to Poisson varieties, which will be used frequently in Hamiltonian reduction. Finally, in 2.4, we introduce the relevant superalgebra structures on superalgebras, especially on Clifford algebras; this part contains the main tools for BRST cohomology.

2.1 Algebraic Geometry

Definition 2.1. Let X be an algebraic variety over $\mathbb{C}$, and let $x \in X$. We say that X is *smooth* at x if the local ring $\mathcal{O}_{X,x}$ is a regular local ring. Equivalently,

$$\dim_{\mathbb{C}}(\mathfrak{m}_x/\mathfrak{m}_x^2) = \dim X,$$

where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$. If X is smooth at every point $x \in X$, then X is called a *smooth algebraic variety over* $\mathbb{C}$.

Definition 2.2. Let X be a smooth algebraic variety over $\mathbb{C}$, and let $x \in X$. Denote by $\mathcal{O}_{X,x}$ the local ring at x, and by $\mathfrak{m}_x \subset \mathcal{O}_{X,x}$ its maximal ideal. The tangent space of X at x is defined as

$$T_x X := \mathrm{Hom}_{\mathbb{C}}\big(\mathfrak{m}_x/\mathfrak{m}_x^2, \mathbb{C}\big).$$

Example 2.3. A finite-dimensional vector space V, regarded as an affine algebraic variety, is smooth, and its tangent space is isomorphic to itself.

Definition 2.4. If $f : X \to Y$ is a morphism between smooth algebraic varieties and $x \in X$, then the differential

$$df_x : T_x X \to T_{f(x)} Y$$

acts on derivations as follows:

$$(df_x(\xi))(g) := \xi(g \circ f), \qquad \xi \in T_x X, \quad g \in \mathcal{O}_{Y,f(x)}.$$

Definition 2.5. Let $f\colon X \to Y$ be a smooth map between algebraic varieties. A point $y \in Y$ is called a regular value of f if, for every $x \in f^{-1}(y)$, the differential

$$df_x\colon T_x X \to T_y Y$$

is surjective.

2.2 Algebraic Groups

Definition 2.6. An *algebraic group* G is a group which is also an algebraic variety, such that the multiplication map $\mu : G \times G \to G$, where $\mu(x, y) = xy$, and the inversion map $i : x \mapsto x^{-1}$ are both morphisms of algebraic varieties, that is,

regular functions. (We may regard the underlying point set of the variety $G \times G$ as the product of point sets [Spr98, exercise 1.5.5].) If its underlying algebraic variety is affine, then G is called a *linear algebraic group*.

Interestingly, [Spr98, theorem 2.3.7]:

Proposition 2.7. *A linear algebraic group G is always isomorphic to a closed subgroup of some GL_n.*

Definition 2.8. For an algebraic group G, its Lie algebra is defined as T_eG, where e is the identity element.

In what follows, unless otherwise specified, all groups G are assumed to be linear algebraic groups. For any linear algebraic group G over a field k, there is a unique maximal connected normal solvable subgroup, called the *radical* of G and denoted by $R(G)$; the subgroup consisting of all unipotent elements in $R(G)$ is a normal closed subgroup of G, and is called the *unipotent radical* of G, denoted by $R_u(G)$ [Spr98, p. 112].

Definition 2.9. If $R_u(G)$ is trivial and $G \neq e$ is connected, then G is called a *reductive group*.

Definition 2.10. If $R(G)$ is trivial and $G \neq e$ is connected, then G is called a *semisimple group*.

Remark 2.11. Semisimple groups are reductive groups.

Example 2.12. The linear algebraic group GL_n is reductive but not semisimple, and its Lie algebra is $\mathfrak{gl}_n$; however, its closed subgroup SL_n is semisimple, and its Lie algebra is $\mathfrak{sl}_n$.

Definition 2.13. Let V be a finite-dimensional vector space.

1. If $n \in \operatorname{End} V$ is nilpotent, then

$$\exp(n) = \sum_{i=0}^{\infty} \frac{n^i}{i!}$$

is a finite sum. It defines an element of $GL(V)$, and $\exp(n)$ is unipotent.

2. If $x \in GL(V)$ is unipotent, then $x - 1$ is nilpotent, and hence

$$\log(x) = \sum_{i=1}^{\infty} \frac{(-1)^{i+1}}{i}(x-1)^i$$

is also a finite sum. It defines a nilpotent element of $\operatorname{End}(V)$.

Proposition 2.14. exp *and* log *give mutually inverse bijections between nilpotent matrices in* $M(n,k)$ *and unipotent matrices in* $GL(n,k)$ *[Hum75, § 15.5 exercise 8].*

From now on, unless otherwise specified, we always assume $k = \mathbb{C}$. For any linear unipotent algebraic group G, consider an isomorphic embedding

$$h: \ G \to GL(V),$$

and take the differential at the identity element to obtain

$$dh: \ \mathfrak{g} \to \operatorname{End}(V).$$

Thus we can define exp and log on G as follows:

$$\exp(g) := h^{-1} \exp(dh(g)) \quad \forall g \in \mathfrak{g};$$
$$\log(x) := dh^{-1} \log(h(x)) \quad \forall x \in G.$$

Remark 2.15. One needs to verify that $\exp(dh(g)) \in h(G)$ and $\log(h(x)) \in dh(\mathfrak{g})$. We only give the proof of the former here.

Proof. Let $Y := dh(X) \in \mathrm{End}(V)$. Since Y is nilpotent, the map

$$\lambda_X : \mathbb{A}^1 \to GL(V), \qquad t \mapsto \exp(tY)$$

is an algebraic morphism. To prove that $\exp(Y) \in h(G)$, it is enough to prove that $\lambda_X(\mathbb{A}^1) \subset h(G)$.

Let $I(h(G)) \subset k[GL(V)]$ be the defining ideal of $h(G)$. For any $f \in I(h(G))$, set

$$p_f(t) := f(\exp(tY)).$$

Then $p_f(t)$ is a polynomial in t.

On $GL(V)$, consider the left-invariant vector field determined by Y:

$$\widetilde{Y}_g := gY \in T_g GL(V).$$

Since $Y \in \mathrm{Lie}(h(G)) = T_I(h(G))$, for any $g \in h(G)$, the left translation L_g preserves $h(G)$, hence

$$\widetilde{Y}_g = d(L_g)_e(Y) \in T_g(h(G)).$$

Therefore

$$\widetilde{Y}(I(h(G))) \subset I(h(G)).$$

On the other hand,

$$p_f'(t) = (\widetilde{Y} f)(\exp(tY)),$$

and by induction we obtain

$$p_f^{(m)}(t) = \big(\widetilde{Y}^m f\big)(\exp(tY)).$$

In particular,

$$p_f^{(m)}(0) = \big(\widetilde{Y}^m f\big)(e).$$

Since $f \in I(h(G))$ and $\widetilde{Y}(I(h(G))) \subset I(h(G))$, we get

$$\widetilde{Y}^m f \in I(h(G)) \qquad (\forall m \geq 0),$$

and hence

$$p_f^{(m)}(0) = 0 \qquad (\forall m \geq 0).$$

Since p_f is a polynomial, it follows that $p_f \equiv 0$. Thus

$$f(\exp(tY)) = 0 \qquad \forall t,\ \forall f \in I(h(G)).$$

Therefore

$$\exp(tY) \in h(G) \qquad \forall t.$$

Taking $t = 1$, we get

$$\exp(dh(X)) = \exp(Y) \in h(G).$$

□

It is easy to see that in this situation exp and log are inverse to each other and give a bijection between the unipotent group and the nilpotent Lie algebra.

Example 2.16. Let G be a connected algebraic group with Lie algebra $\mathfrak{g}$. Consider the adjoint orbit map

$$\mu_f : G \to \mathfrak{g}, \qquad g \mapsto \mathrm{Ad}(g)f.$$

Since $\mathfrak{g}$ is a vector space, we make the identification

$$T_f\mathfrak{g} \cong \mathfrak{g}.$$

Also,

$$T_1 G \cong \mathfrak{g}.$$

For $x \in \mathfrak{g}$, take the one-parameter subgroup $g(t) = \exp(tx)$. Then

$$d\mu_f|_1(x) = \frac{d}{dt}\bigg|_{t=0} \mathrm{Ad}(\exp(tx))f = \frac{d}{dt}\bigg|_{t=0} e^{t\,\mathrm{ad}\,x}(f) = [x, f].$$

Therefore the tangent space to the adjoint orbit $G \cdot f$ at f is

$$T_f(G \cdot f) = \mathrm{Im}(d\mu_f|_1) = [\mathfrak{g}, f] = (\mathrm{ad}\, f)(\mathfrak{g}).$$

2.3 Symplectic Geometry

Let M be a closed subgroup of a semisimple algebraic group G. We first introduce the coadjoint action. The conjugation action of M on G is defined by

$$\mathrm{Int} : M \to \mathrm{Aut}(G), \qquad m \mapsto \mathrm{Int}_m,$$

where

$$\mathrm{Int}_m(g') = mg'm^{-1}.$$

Taking the differential of Int_m at the identity element gives the adjoint action

$$\mathrm{Ad}_m : \mathfrak{g} \to \mathfrak{g}, \qquad g \mapsto \left.\frac{d}{dt}\right|_{t=0} \left(m \exp(tg) m^{-1}\right).$$

Taking the dual map gives

$$\mathrm{Ad}_m^* : \mathfrak{g}^* \to \mathfrak{g}^*.$$

Therefore M acts on $\mathfrak{g}^*$ by the coadjoint action:

$$M \to \mathrm{GL}(\mathfrak{g}^*), \qquad m \mapsto \mathrm{Ad}_m^*.$$

Taking the differential at the identity element again gives

$$\mathfrak{m} \to \mathrm{End}(\mathfrak{g}^*), \qquad m' \mapsto \mathrm{ad}_{m'}^*.$$

Proposition 2.17 (*Kirillov–Kostant–Souriau* symplectic structure)**.** *Every coadjoint orbit $\mathcal{O} \subset \mathfrak{g}^*$ has the following natural symplectic structure [CG10, § 1].*

Take a point $\alpha \in \mathcal{O} \subset \mathfrak{g}^*$. We have a skew-symmetric form on $T_\alpha \mathcal{O} \simeq \mathfrak{g}/\mathfrak{g}^\alpha$, where $\mathfrak{g}^\alpha = \mathrm{Lie}\, G^\alpha$.

We first define a skew-symmetric form

$$\omega_\alpha \colon \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}, \qquad \omega_\alpha(x, y) \mapsto \alpha([x, y]).$$

It descends to $\mathfrak{g}/\mathfrak{g}^\alpha$, that is,

$$\omega_\alpha(x, y) = 0 \quad \forall\, y \in \mathfrak{g} \iff x \in \mathfrak{g}^\alpha.$$

The form ω_α is also nondegenerate, and hence ω_α defines a symplectic form on $\mathcal{O}$. The assignment $\alpha \mapsto \omega_\alpha$ gives a closed 2-form ω on $\mathcal{O}$, which is a symplectic form.

Let A be a commutative, associative, unital $\mathbb{C}$-algebra, with multiplication $\cdot\colon A \times A \to A$.

Definition 2.18. A commutative algebra $(A, \cdot)$ equipped with an additional $\mathbb{C}$-bilinear antisymmetric structure $\{\,,\}\colon A \times A \to A$ is called a *Poisson algebra* if it satisfies the following conditions:

1. A is a Lie algebra with respect to $\{\,,\}$;

2. Leibniz rule:

$$\{a, b \cdot c\} = \{a, b\} \cdot c + b \cdot \{a, c\}, \qquad \forall a, b, c \in A.$$

The Lie bracket $\{\,,\}$ is called a *Poisson bracket* on A. We say that $\{\,,\}$ gives a *Poisson structure* on the commutative algebra $(A, \cdot)$.

We now derive from a symplectic structure a Poisson structure called the Lie–Poisson bracket. Let (M, ω) be a symplectic variety. The nondegenerate 2-form ω gives a canonical isomorphism $TM \simeq T^*M$. Define the unique $\mathbb{C}$-linear map

$$\mathcal{O}(M) \to \{\text{vector fields on } M\}, \quad f \mapsto \xi_f,$$

such that

$$\omega(\,\cdot\,, \xi_f) = df.$$

Note that for any vector field η and any function f, by the definition of ξ_f we have

$$\omega(\xi_f, \eta) = -\eta f.$$

We define a bracket on $\mathcal{O}(M)$ by any of the following equivalent expressions:

$$\{f, g\} = \omega(\xi_f, \xi_g) = -\xi_g f = \xi_f g.$$

Definition 2.19 (Reduced scheme)**.** A scheme $(X, \mathcal{O}_X)$ is called *reduced* if, for every open set $U \subseteq X$, the ring $\mathcal{O}_X(U)$ has no nilpotent elements.

Definition 2.20. An *affine Poisson scheme* (*affine Poisson variety*) is an affine scheme $X = \operatorname{Spec} A$ ($= \operatorname{Specm} A$), where A is a Poisson algebra. A *Poisson scheme* (*Poisson variety*) is a scheme (reduced scheme) whose structure sheaf $\mathcal{O}_X$ is a sheaf of Poisson algebras.

Example 2.21. Let $\mathfrak{g}$ be a finite-dimensional Lie algebra. Then its dual space $\mathfrak{g}^*$ is naturally a Poisson variety. Indeed, as an affine space,

$$\mathfrak{g}^* = \operatorname{Spec} \mathbb{C}[\mathfrak{g}^*],$$

and

$$\mathbb{C}[\mathfrak{g}^*] \cong S(\mathfrak{g}).$$

For $x, y \in \mathfrak{g} \subset S(\mathfrak{g})$, define

$$\{x, y\} := [x, y]_{\mathfrak{g}},$$

and extend it uniquely to all of $S(\mathfrak{g})$ by the Leibniz rule. Then $S(\mathfrak{g}) \cong \mathbb{C}[\mathfrak{g}^*]$ becomes a Poisson algebra. Therefore $\mathfrak{g}^*$ is an affine Poisson variety.

Definition 2.22. A Hamiltonian flow from x to x' is a smooth curve $\gamma\colon [0, 1] \to X$, defined on an open neighborhood in $\mathbb{C}$ containing $[0, 1]$, satisfying $\gamma(0) = x$ and $\gamma(1) = x'$, and such that γ is an integral curve of some Hamiltonian vector field $\xi_f = \{f, \cdot\}$, where $f \in \mathcal{O}(X)$ is defined on an open neighborhood of $\gamma([0, 1])$.

2.4 Clifford Algebras

We introduce superalgebras, especially Clifford algebras.

Definition 2.23. A superspace is a $\mathbb{Z}_2$-graded $\mathbb{C}$-linear space E, that is, $E = E_{\bar{0}} \oplus E_{\bar{1}}$. Elements of $E_{\bar{0}}$ are called *even*, and elements of $E_{\bar{1}}$ are called *odd*. A superspace which is also an algebra is called a *superalgebra*.

Definition 2.24. Let $\mathcal{A} = \mathcal{A}_{\bar{0}} \oplus \mathcal{A}_{\bar{1}}$ be a superspace. If a bilinear map

$$[\cdot, \cdot] : \mathcal{A} \times \mathcal{A} \to \mathcal{A}$$

is given on $\mathcal{A}$ such that, for any homogeneous elements $x, y, z \in \mathcal{A}$,

$$\begin{aligned} [x, y] &= -(-1)^{|x||y|}[y, x], \\ [x, [y, z]] &= -[z, [x, y]] - (-1)^{|x||y|+1}[y, [x, z]] \quad \text{(Jacobi super-identity)}, \end{aligned}$$

where $|\cdot|$ is the degree operator, then $(\mathcal{A}, [\cdot, \cdot])$ is called a *Lie superalgebra*.

Remark 2.25. Every superalgebra $\mathcal{A}$ can be given a natural Lie superalgebra structure by defining

$$[x, y] := xy - (-1)^{|x||y|}yx.$$

If, for any homogeneous elements $x, y \in \mathcal{A}$, one has

$$[x, y] = 0,$$

that is,

$$xy = (-1)^{|x||y|}yx,$$

then $\mathcal{A}$ is called *supercommutative*.

Definition 2.26. Let $\mathcal{A} = \mathcal{A}_{\bar{0}} \oplus \mathcal{A}_{\bar{1}}$ be a supercommutative superalgebra. Suppose that a bilinear map

$$\{\cdot, \cdot\} : \mathcal{A} \times \mathcal{A} \to \mathcal{A}$$

is also given on $\mathcal{A}$ such that:

1. $(\mathcal{A}, \{\cdot,\cdot\})$ is a Lie superalgebra;

2. for any homogeneous elements $x, y, z \in \mathcal{A}$, the bracket satisfies the super Leibniz rule with respect to multiplication:

$$\{x, yz\} = \{x, y\}z + (-1)^{|x||y|}y\{x, z\},$$

that is, $\{x, \cdot\} : \mathcal{A} \to \mathcal{A}$ is a superderivative,

then $(\mathcal{A}, \cdot, \{\cdot,\cdot\})$ is called a *Poisson superalgebra*.

Example 2.27. Let E be a $\mathbb{C}$-linear space. The *exterior algebra* $\wedge E$ is the quotient of the tensor algebra $T(E) = \bigoplus_{k\in\mathbb{Z}} T(E)^k$ by the two-sided ideal $I(E)$ generated by all elements of the form $v \otimes w + w \otimes v$, where $v, w \in E$. Here $T^k(E) = E \otimes \cdots \otimes E$ is the k-fold tensor product. Multiplication in $\wedge E$ is usually denoted by $v \wedge w$. Since $I(E)$ is graded, the exterior algebra has a $\mathbb{Z}$-grading

$$\wedge(E) = \bigoplus_{k\in\mathbb{Z}} \wedge^k(E).$$

We regard $\wedge(E)$ as a graded superalgebra, where the $\mathbb{Z}_2$-grading is obtained by reducing the $\mathbb{Z}$-grading modulo 2.

Definition 2.28. The Clifford algebra $Cl(E, B)$ is the quotient of $T(E)$ by the two-sided ideal $\mathcal{I}(E, B)$ generated by all elements of the form

$$v \otimes w + w \otimes v - B(v, w)1, \quad v, w \in E,$$

where $B : E \times E \to E$ is a symmetric bilinear form (usually assumed to be nondegenerate).

Let us regard $T(E) = \bigoplus_{k\in\mathbb{Z}} T^k(E)$ as a filtered superalgebra, whose $\mathbb{Z}_2$-grading and filtration are both inherited from the $\mathbb{Z}$-grading. Since the element $v \otimes w + w \otimes v - B(v, w)1$ is even and has filtration degree 2, the ideal $\mathcal{I}(E, B)$

is a filtered supersubspace of $T(E)$. Therefore $Cl(E,B)$ inherits the structure of a filtered superalgebra.

In the decomposition

$$Cl(E,B) = Cl(E,B)^{\bar{0}} \oplus Cl(E,B)^{\bar{1}},$$

the two direct summands are spanned respectively by products $v_1 \dots v_k$ with k even and odd. We always regard $Cl(E,B)$ as a filtered superalgebra. Thus the defining relation of the Clifford algebra becomes

$$[v,w] = vw + wv = B(v,w), \qquad v,w \in E.$$

We define the map:

$$q\colon \wedge(E) \to Cl(E,B), \quad v_1 \wedge \dots \wedge v_k \mapsto \sum_{\sigma \in \mathfrak{S}_k} \operatorname{sgn}(\sigma) v_{\sigma 1} \dots v_{\sigma k}, \tag{2.i}$$

where $\mathfrak{S}_k$ is the symmetric group of degree k. It is clear that this map is an isomorphism of superspaces. Its inverse is called the *symbol map*. From [Mei13, § 2.2.5], we then have

Proposition 2.29. *The symbol map $\sigma\colon Cl(E,B) \to \wedge(E)$ induces an isomorphism of graded superalgebras*

$$\operatorname{gr} Cl(E,B) \xrightarrow{\sim} \wedge(E).$$

$\operatorname{gr} Cl(E,B)$ inherits a Poisson superalgebra structure, and the symbol map is an isomorphism of graded Poisson superalgebras. The Poisson bracket on $\wedge(E)$ can be described by

$$\{v,w\} = B(v,w), \quad v,w \in E = \wedge^1(E).$$

Let $(\mathcal{A}, \{,\}_{\mathcal{A}})$ and $(\mathcal{B}, \{,\}_{\mathcal{B}})$ be two Poisson algebras. Consider the tensor product of associative algebras $\mathcal{A} \otimes_{\mathbb{C}} \mathcal{B}$ and equip it with the following Poisson bracket:

$$\{x_1 \otimes y_1, x_2 \otimes y_2\} = \{x_1, x_2\}_A \otimes y_1 \cdot y_2 + x_1 \cdot x_2 \otimes \{y_1, y_2\}_B,$$

where $x_1, x_2 \in A$, $y_1, y_2 \in B$. It is easy to verify that $A \otimes_{\mathbb{C}} B$ is a Poisson algebra. In addition, since $1 \otimes B$ is isomorphic to B as a Poisson algebra, we may write $B \subset A \otimes_{\mathbb{C}} B$ (here we use the fact that $\{1, x\} = 0$ for all $x \in A$).

This observation can be easily generalized to the case of Clifford algebras. In that case, the Poisson bracket becomes

$$\{x_1 \otimes y_1, x_2 \otimes y_2\} = (-1)^{|x_2||y_1|}\{x_1, x_2\}_A \otimes y_1 \cdot y_2 + (-1)^{|x_2|(|x_1|+|y_1|)} x_1 \cdot x_2 \otimes \{y_1, y_2\}_B.$$

Chapter 3

Slodowy Slices and Their Poisson Structure

In this chapter, we will study in depth the Poisson structure on Slodowy slices. In Section 3.1, we use the Jacobson–Morozov theorem to define Slodowy slices and prove their transversality. In Section 3.2, we investigate some elementary properties of the nilpotent cone. In Section 3.3, we introduce Slodowy slices in the dual space and describe the Poisson structure on them in two ways: by restricting the original Poisson structure to symplectic leaves, and via Hamiltonian reduction. We mainly follow the notation in [Slo80], [AM16], and [GG02].

3.1 The Jacobson–Morozov Theorem and Slodowy Slices

Let $\mathfrak{sl}_2 = ke + kh + kh$, where the elements are

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

and it's easy to see they satisfy the following commutation relations:

$$[h, e] = 2e,\ [h, f] = -2f,\ [e, f] = h.$$

Definition 3.1. For a nilpotent element e in $\mathfrak{g}$, if there is a triple (e, h, f) in $\mathfrak{g}$ and a Lie algebra homomorphism $\rho : \mathfrak{sl}_2 \to \mathfrak{g}$ such that $\rho(E) = e$, $\rho(H) = h$, and $\rho(F) = f$, then it is called an $\mathfrak{sl}_2$–triple for the nilpotent element e.

Definition 3.2. A representation $\rho : \mathfrak{sl}_2 \to \mathfrak{gl}(V)$ of $\mathfrak{sl}_2$ on a finite-dimensional vector space V is called *good* if either $char(k) = 0$, or $char(k) = p > 0$ and $\rho(E)^{p-1} = \rho(F)^{p-1} = 0$.

Let the d-dimensional vector space V_d be spanned by basis vectors $v_1, \ldots, v_d$. If char$(k) = p > 0$, then assume $d < p$; otherwise no restriction is imposed on d. One can define a good, d-dimensional, irreducible representation of $\mathfrak{sl}_2$ by

$$\begin{aligned}
\rho_d(E)v_1 &= \rho_d(F)v_d = 0 \\
\rho_d(E)v_{i+1} &= \overline{i(d-i)}\, v_i \quad \text{for } 1 \le i \le d-1 \\
\rho_d(F)v_i &= v_{i+1} \quad \text{for } 1 \le i \le d-1 \\
\rho_d(H)v_i &= \overline{(d-2i+1)}\, v_i
\end{aligned}$$

(where, for $n \in \mathbb{Z}$, $\overline{n}$ denotes the residue class modulo p.)

The following theorem is the key to defining Slodowy slices. Let $\mathfrak{g}$ be the Lie algebra of a semisimple algebraic group G (of type Δ)[1], and let $x \in \mathcal{N}(\mathfrak{g})$ be a nilpotent element of $\mathfrak{g}$.

The *Coxeter number* Cox(G) or Cox(Δ) of G or Δ is defined as $m+1$, where m is the height of the highest root in the root system of G. The values of Cox(G) are as follows [Slo80, § 7]:

Next we have the famous Jacobson–Morozov theorem [Slo80, § 7.2]:

[1] We know that $\mathfrak{g}$ is semisimple; see [Hum75, § 13.5]

Type of G	A_r	B_r	C_r	D_r	E_6	E_7	E_8	F_4	G_2
$\mathrm{Cox}(G)$	$r+1$	$2r$	$2r$	$2r-2$	12	18	30	12	6

Theorem 3.3 (Jacobson–Morozov). *Assume* $\mathrm{char}(k) = 0$ *or* $> 4\,\mathrm{Cox}(G) - 2$. *For every nilpotent element* $e \in \mathfrak{g}$, *there exists an* $\mathfrak{sl}_2$*–triple, and the corresponding representation* $\rho : \mathfrak{sl}_2 \to \mathfrak{g}$, *composed with the adjoint representation* $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$, *is good.*

In this paper, we mainly focus on the Lie algebra $\mathfrak{sl}_n$ corresponding to the group SL_n. Every nilpotent matrix in $\mathfrak{sl}_n$ is conjugate to a Jordan block diagonal matrix. Therefore, nilpotent orbits in $\mathfrak{g}$ are parametrized by $\mathcal{P}(n)$. Thus nilpotent orbits in $\mathfrak{sl}_n$ correspond to the set $\mathcal{P}(n)$ of partitions of n.

Remark 3.4. Let $n \in \mathbb{Z}_{>0}$. We write an element $\boldsymbol{\lambda}$ of the set of partitions of n, regarded as an element of $\mathcal{P}(n)$, as a decreasing sequence $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_s)$, omitting the zero terms. Thus,

$$\lambda_1 \geq \cdots \geq \lambda_s \geq 1 \quad \text{and} \quad \lambda_1 + \cdots + \lambda_s = n.$$

We denote the dual partition of a partition $\boldsymbol{\lambda} \in \mathcal{P}(n)$ by ${}^t\boldsymbol{\lambda}$; its Young tableau is the transpose of the Young tableau of $\boldsymbol{\lambda}$.

Example 3.5. Let $\mathfrak{g} = \mathfrak{sl}_n$, and set

$$e := J_n, \qquad h := \sum_{i=1}^{n} (n+1-2i) e_{i,i}, \qquad f := \sum_{i=1}^{n-1} i(n-i) e_{i+1,i}.$$

Then (e, h, f) is an $\mathfrak{sl}_2$–triple. From this observation, we can immediately construct an $\mathfrak{sl}_2$–triple for any Jordan normal form $\mathrm{diag}(J_{\lambda_1}, \ldots, J_{\lambda_n})$, where $(\lambda_1, \ldots, \lambda_n) \in \mathcal{P}(n)$. For the partition $(3, 1)$, we obtain

$$e = E_{12} + E_{23}, \qquad h = \mathrm{diag}(2, 0, -2, 0), \qquad f = 2E_{21} + 2E_{32}.$$

One can directly verify that

$$[h, e] = 2e, \qquad [e, f] = h, \qquad [h, f] = -2f.$$

Next we state a theorem from [Jac58]:

Theorem 3.6. *Every good representation ρ is completely reducible and decomposes into a direct sum of irreducible good representations of type ρ_d, where $d \in \mathbb{N}^+$, and if* $\mathrm{char}(k) = p > 0$*, then one further requires $d < p$. In particular, for every natural number $d \in \mathbb{N}^+$, with the additional requirement $d < p$ when* $\mathrm{char}(k) = p > 0$*, there exists exactly one irreducible representation of dimension d.*

Definition 3.7. Let V be a smooth algebraic variety, let $\mathcal{O} \subset V$ be a smooth subvariety, and let $y \in \mathcal{O}$. If a subvariety $S \subset V$ through y satisfies

$$T_y V = T_y \mathcal{O} \oplus T_y S,$$

then S is called a *transversal slice to $\mathcal{O}$ at y*.

Let (e, h, f) be an $\mathfrak{sl}_2$–triple in $\mathfrak{g}$. Consider the orbit map

$$\mu_f : G \to \mathfrak{g}, \qquad g \mapsto \mathrm{Ad}(g) f.$$

By Example 2.4, the tangent space to the adjoint orbit $G \cdot f$ at f is

$$T_f(G \cdot f) = (\mathrm{ad}\, f)(\mathfrak{g}).$$

By Theorem 3.6, the adjoint representation of the $\mathfrak{sl}_2$–triple decomposes as

$$\mathfrak{g} = \bigoplus_{j=1}^{s} E_j,$$

where each E_j is an irreducible $\mathfrak{sl}_2$-module of dimension $n_j + 1$. Write

$$E_j = \bigoplus_{m=-n_j}^{n_j} E_j(m)$$

for its weight-space decomposition. Then each $E_j(m)$ is one-dimensional, and for every

$$m \leq n_j - 2$$

the map ad f restricts to an isomorphism

$$\operatorname{ad} f : E_j(m+2) \xrightarrow{\sim} E_j(m).$$

Therefore,

$$(\operatorname{ad} f)(E_j) = \bigoplus_{m=-n_j}^{n_j-2} E_j(m),$$

so the highest weight space $E_j(n_j)$ is a complement to $(\operatorname{ad} f)(E_j)$ in E_j. Since

$$E_j(n_j) = \ker(\operatorname{ad} e) \cap E_j,$$

we obtain

$$\ker(\operatorname{ad} e) = \bigoplus_{j=1}^{s} E_j(n_j).$$

Therefore,

$$\mathfrak{g} = (\operatorname{ad} f)(\mathfrak{g}) \oplus \ker(\operatorname{ad} e).$$

Definition 3.8. The Slodowy slice through f is the affine subspace

$$\mathcal{S}_f := f + \ker(\operatorname{ad} e) = f + \mathfrak{g}^e,$$

Proposition 3.9. *The Slodowy slice $\mathcal{S}_f$ is transversal to the adjoint orbit $G \cdot f$ at the point f. More precisely,*

$$T_f\mathfrak{g} = T_f(G \cdot f) \oplus T_f(\mathcal{S}_f).$$

Proof. Since $\mathcal{S}_f = f + \mathfrak{g}^e$ is an affine subspace of $\mathfrak{g}$, its tangent space at f is

$$T_f(\mathcal{S}_f) = \mathfrak{g}^e = \ker(\operatorname{ad} e).$$

On the other hand,

$$T_f(G \cdot f) = (\operatorname{ad} f)(\mathfrak{g}).$$

Using the decomposition above,

$$\mathfrak{g} = (\operatorname{ad} f)(\mathfrak{g}) \oplus \ker(\operatorname{ad} e),$$

we obtain

$$T_f\mathfrak{g} = \mathfrak{g} = T_f(G \cdot f) \oplus T_f(\mathcal{S}_f).$$

Therefore $\mathcal{S}_f$ is transversal to the orbit $G \cdot f$ at f. $\square$

In particular, $\mathcal{S}_f$ defines a transversal slice to the adjoint orbit of f at the point f.

Example 3.10. Let $Y = (y_{ij}) \in \mathfrak{sl}_4$. Solving the equation

$$[Y, e] = 0, \qquad e = E_{12} + E_{23},$$

we obtain

$$Y = \begin{pmatrix} a & b & c & u \\ 0 & a & b & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & v & -3a \end{pmatrix}.$$

Therefore

$$\mathfrak{g}^e = \left\{ \begin{pmatrix} a & b & c & u \\ 0 & a & b & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & v & -3a \end{pmatrix} \middle| \; a, b, c, u, v \in \mathbb{C} \right\}.$$

Hence

$$\mathcal{S}_f = f + \mathfrak{g}^e = \left\{ \begin{pmatrix} a & b & c & u \\ 2 & a & b & 0 \\ 0 & 2 & a & 0 \\ 0 & 0 & v & -3a \end{pmatrix} \middle| \; a, b, c, u, v \in \mathbb{C} \right\}.$$

Thus, as an affine algebraic variety,

$$\mathcal{S}_f \cong \mathbb{A}^5,$$

and its coordinate ring is

$$\mathbb{C}[\mathcal{S}_f] \cong \mathbb{C}[a, b, c, u, v].$$

3.2 Nilpotency

Definition 3.11. The set of all nilpotent elements in $\mathfrak{g}$ is called the *nilpotent cone* of $\mathfrak{g}$, and is denoted by $\mathcal{N}$.

For $e \in \mathfrak{g}$, we denote its G-orbit under the adjoint action by $G.e$. The nilpotent cone is the union of finitely many nilpotent G-orbits. An element $x \in \mathfrak{g}$ is called *regular* if its centralizer $\mathfrak{g}^x$ has the minimal possible dimension, namely the rank r of $\mathfrak{g}$. The orbit of maximal dimension, denoted by $\mathcal{O}_{\text{reg}}$, is the set of all regular nilpotent elements.

Notice that if $\mathfrak{g}$ is a simple Lie algebra of matrices, then by the properties of Jordan decomposition, $\mathcal{N}$ can be identified with the set of nilpotent matrices [Hum72, § 6.3].

Example 3.12. In $\mathfrak{sl}_n$, regular nilpotent elements are the conjugacy class of the nilpotent Jordan block of size n:

$$e = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

In $\mathfrak{gl}_n$, the matrices commuting with e are precisely the polynomials in e:

$$a_0 I + a_1 e + \cdots + a_{n-1} e^{n-1}.$$

Thus their dimension is n.

After taking the trace-zero part in $\mathfrak{sl}_n$, the generator I is removed. Therefore

$$\dim \mathfrak{g}^e = n - 1 = \operatorname{rank} \mathfrak{sl}_n.$$

Next, in $\mathcal{N} \setminus \mathcal{O}_{\text{reg}}$ there is a unique orbit of maximal dimension. It is called the *subregular nilpotent orbit* of $\mathfrak{g}$, and is denoted by $\mathcal{O}_{\text{subreg}}$.

Example 3.13. Consider the nilpotent Jordan form corresponding to the partition $(n-1, 1)$, namely one Jordan block of size $(n-1)$ and one Jordan block of size 1. Denote it by X.

$$\mathfrak{g}^X = \{Y \in \mathfrak{sl}_n : [X, Y] = 0\}.$$

Let

$$Y = \begin{pmatrix} A & b \\ c & d \end{pmatrix},$$

where $A \in M_{n-1}$, $b \in M_{n-1,1}$, $c \in M_{1,n-1}$, and $d \in \mathbb{C}$.

$$X = \begin{pmatrix} J & 0 \\ 0 & 0 \end{pmatrix}, \quad J := J_{n-1},$$

where J_{n-1} is the nilpotent Jordan block of size $(n-1)$. Then

$$XY = \begin{pmatrix} JA & Jb \\ 0 & 0 \end{pmatrix}, \quad YX = \begin{pmatrix} AJ & 0 \\ cJ & 0 \end{pmatrix}.$$

Therefore

$$JA = AJ, \quad Jb = 0, \quad cJ = 0.$$

Hence

$$c = (0, \ldots, 0, c_{n-1}), \quad b = (b_1, 0, \ldots, 0)^T, \quad A = a_0 I + a_1 J + \cdots + a_{n-2} J^{n-2}.$$

Since $\operatorname{tr} Y = 0$, d is uniquely determined by A. Thus the dimension is uniquely determined by the above description, namely

$$1 + 1 + n - 1 = n - 1 + 2 = \operatorname{rank} \mathfrak{sl}_n + 2.$$

In particular, in most of this section we focus on the subregular case of $\mathfrak{sl}_4$, namely the partition $(3, 1)$ of 4. In the following examples, we will no longer distinguish between $\mathfrak{g}$ and $\mathfrak{sl}_4$.

Example 3.14. We denote the nilpotent orbit in $\mathfrak{sl}_n$ corresponding to $\boldsymbol{\lambda}$ by $\mathcal{O}_{\boldsymbol{\lambda}}$. Then $\mathcal{O}_{\boldsymbol{\lambda}}$ is represented by the standard Jordan form $\operatorname{diag}(J_{\lambda_1}, \ldots, J_{\lambda_s})$, where J_k is the Jordan block of size k. If we write ${}^t\boldsymbol{\lambda} = (d_1, \ldots, d_t)$, then one can prove that, for some $e \in \mathcal{O}_{\boldsymbol{\lambda}}$,

$$\dim \mathfrak{g}^e = \sum_i d_i^2 - 1.$$

Therefore

$$\dim \mathcal{O}_{\boldsymbol{\lambda}} = n^2 - \sum_{i=1}^{t} d_i^2.$$

For $\boldsymbol{\lambda} = (3, 1)$ in $\mathfrak{sl}_4$, since ${}^t(3, 1) = (2, 1, 1)$, we get

$$\dim \mathcal{O}_{(3,1)} = 4^2 - (2^2 + 1^2 + 1^2) = 16 - 6 = 10.$$

Thus

$$\dim \mathfrak{g}^e = \dim \mathfrak{sl}_4 - \dim \mathcal{O}_{(3,1)} = 15 - 10 = 5.$$

By the Jacobson–Morozov theorem 3.3, we obtain an $\mathfrak{sl}_2$–triple e, f, h.

Definition 3.15. Since the eigenvalues of $\operatorname{ad} h$ are all integers, we obtain a $\frac{1}{2}\mathbb{Z}$-grading of $\mathfrak{g}$ defined by h. It is called the *Dynkin grading* associated with h:

$$\mathfrak{g} = \bigoplus_{j \in \frac{1}{2}\mathbb{Z}} \mathfrak{g}_j, \qquad \mathfrak{g}_j := \{x \in \mathfrak{g} \mid [h, x] = 2jx\}.$$

If $\mathfrak{g}_i = \{0\}$ for all half-integers i, then e is called an *even nilpotent element*. A nilpotent orbit consisting of even nilpotent elements is called an *even nilpotent orbit*.

Example 3.16. Since

$$h = \text{diag}(2, 0, -2, 0),$$

let

$$w_1 = 2, \qquad w_2 = 0, \qquad w_3 = -2, \qquad w_4 = 0.$$

Then

$$[h, E_{ij}] = (w_i - w_j)E_{ij}.$$

Therefore

$$\mathfrak{g}_2 = \mathbb{C}E_{13},$$

$$\mathfrak{g}_1 = \text{span}\{E_{12}, E_{14}, E_{23}, E_{43}\},$$

$$\mathfrak{g}_0 = \left\{ \begin{pmatrix} a & 0 & 0 & 0 \\ 0 & b & 0 & u \\ 0 & 0 & c & 0 \\ 0 & v & 0 & d \end{pmatrix} \middle| \; a + b + c + d = 0 \right\},$$

$$\mathfrak{g}_{-1} = \text{span}\{E_{21}, E_{41}, E_{32}, E_{34}\}, \qquad \mathfrak{g}_{-2} = \mathbb{C}E_{31}.$$

In particular, this nilpotent element is *even*.

Below we will often identify $\mathfrak{g}$ with its dual $\mathfrak{g}^*$ by means of a non-degenerate invariant bilinear form $(\ |\ )$. In particular, in the semisimple type A case, this non-degenerate bilinear form can be taken to be the Killing form. Let (e, h, f) be an $\mathfrak{sl}_2$–triple in $\mathfrak{g}$, and let $\chi = (f|\cdot)$ be the linear functional associated with f. Then we have the following proposition.

Proposition 3.17. *The skew-symmetric bilinear form*

$$\omega_\chi : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}, \qquad (x, y) \mapsto (f|[x, y]),$$

restricted to $\mathfrak{g}_{\frac{1}{2}} \times \mathfrak{g}_{\frac{1}{2}}$*, is non-degenerate.*

Proof. It is easy to see that $\operatorname{ad} f : \mathfrak{g}_{\frac{1}{2}} \to \mathfrak{g}_{-\frac{1}{2}}$ is a bijection, whose inverse is $\operatorname{ad} e$. Hence

$$\omega_\chi(x, y) = (f \mid [x, y]) = ([f, x] \mid y),$$

because the bilinear form is invariant under the Lie bracket. Using the fact that $(\ \mid\ )$ is non-degenerate on $\mathfrak{g}_{\frac{1}{2}} \times \mathfrak{g}_{-\frac{1}{2}}$, if $\omega_\chi(x, y) = 0$ for all y, then, since the above map $\operatorname{ad} f$ is injective, we get $[f, x] = 0$, and hence $x = 0$. Therefore ω_χ is non-degenerate on $\mathfrak{g}_{\frac{1}{2}}$. $\square$

The form ω_χ is called the *Kirillov form associated with* f. Let $\mathcal{L}$ be a Lagrangian subspace of $\mathfrak{g}_{\frac{1}{2}}$, that is, $\omega_\chi(\mathcal{L}, \mathcal{L}) = 0$ and $\dim \mathcal{L} = \frac{1}{2} \dim \mathfrak{g}_{\frac{1}{2}}$. Define

$$\mathfrak{m} = \mathfrak{m}_{\chi,\mathcal{L}} := \mathcal{L} \oplus \bigoplus_{j > \frac{1}{2}} \mathfrak{g}_j. \tag{3.i}$$

Then $\mathfrak{m}$ is an ad-nilpotent $\operatorname{ad} h$-graded subalgebra of $\mathfrak{g}$, because $[\mathfrak{g}_i, \mathfrak{g}_j] \subset \mathfrak{g}_{i+j}$. Moreover, since $(\mathfrak{g}_i \mid \mathfrak{g}_j) = 0$ unless $i + j = 0$, we have $\chi([\mathfrak{m}, \mathfrak{m}]) = (f|[\mathfrak{m}, \mathfrak{m}]) = 0$, that is, χ is a character of $\mathfrak{m}$.

Example 3.18. In our case,

$$\chi = (f \mid \cdot), \qquad f = 2E_{21} + 2E_{32}.$$

Since it is well known that $k(A, B) := \operatorname{tr}(\operatorname{ad} A \cdot \operatorname{ad} B) = 2n \operatorname{tr}(AB)$, it is enough to compute the trace pairing $\chi := tr(A\cdot)$:

$$\chi(E_{12}) = 2, \qquad \chi(E_{23}) = 2, \qquad \chi(E_{14}) = 0, \qquad \chi(E_{43}) = 0, \qquad \chi(E_{13}) = 0.$$

Moreover, since

$$\mathfrak{g}_{1/2} = 0,$$

we have $\mathcal{L} = 0$, and hence

$$\mathfrak{m} = \mathfrak{g}_1 \oplus \mathfrak{g}_2.$$

Therefore

$$\mathfrak{m} = \operatorname{span}\{E_{12}, E_{14}, E_{23}, E_{43}, E_{13}\}.$$

3.3 The Poisson structure on $\mathcal{S}_f$

First, we consider the realization of the Slodowy slice in the dual space $\mathfrak{g}^*$. Let

$$\chi := (f \mid \cdot) \in \mathfrak{g}^*.$$

We define

$$\mathcal{S}_f := \chi + (\mathfrak{g}^f)^* \subset \mathfrak{g}^*.$$

We now explain that, by using the invariant bilinear form, this affine subspace is naturally identified with the original definition

$$\mathcal{S}_f = f + \mathfrak{g}^e.$$

Proposition 3.19. *Let*

$$\kappa : \mathfrak{g} \xrightarrow{\sim} \mathfrak{g}^*, \qquad x \mapsto (x \mid \cdot)$$

be the isomorphism induced by the invariant bilinear form. Then

$$\kappa\big(f + \mathfrak{g}^e\big) = \chi + (\mathfrak{g}^f)^*.$$

Equivalently,

$$\mathcal{S}_f \cong f + \mathfrak{g}^e.$$

Proof. Since $\kappa(f) = \chi$, it is enough to prove that the restriction of κ induces an isomorphism

$$\mathfrak{g}^e \xrightarrow{\sim} (\mathfrak{g}^f)^*.$$

Consider the linear map

$$\varphi : \mathfrak{g}^e \to (\mathfrak{g}^f)^*, \qquad x \mapsto (x \mid \cdot)\big|_{\mathfrak{g}^f}.$$

We first prove that φ is injective. Suppose that $x \in \mathfrak{g}^e$ satisfies

$$(x \mid y) = 0 \qquad \text{for all } y \in \mathfrak{g}^f.$$

By the decomposition of $\mathfrak{g}$ as an $\mathfrak{sl}_2$-module, as in Definition 3.9, we have

$$\mathfrak{g} = [\mathfrak{g}, f] \oplus \mathfrak{g}^e.$$

On the other hand, by the invariance of $(\cdot \mid \cdot)$, for any $z \in \mathfrak{g}$ and any $y \in \mathfrak{g}^f$,

$$([z, f] \mid y) = (z \mid [f, y]) = 0,$$

because $[f, y] = 0$. Therefore

$$[\mathfrak{g}, f] \subseteq (\mathfrak{g}^f)^\perp.$$

Now

$$\dim[\mathfrak{g}, f] = \dim \mathfrak{g} - \dim \mathfrak{g}^f = \dim(\mathfrak{g}^f)^\perp,$$

so

$$[\mathfrak{g}, f] = (\mathfrak{g}^f)^\perp.$$

Thus $x \in \mathfrak{g}^e \cap (\mathfrak{g}^f)^\perp = \mathfrak{g}^e \cap [\mathfrak{g}, f]$. Since

$$\mathfrak{g} = [\mathfrak{g}, f] \oplus \mathfrak{g}^e,$$

this intersection is zero, and hence $x = 0$. Therefore φ is injective.

Next we compare dimensions. Since we have proved in Definition 3.9 that $\ker(\operatorname{ad} e) = \bigoplus_{j=1}^s E_j(n_j)$, and similarly $\ker(\operatorname{ad} f) = \bigoplus_{j=1}^s E_j(-n_j)$, we have

$$\dim \mathfrak{g}^e = \dim \mathfrak{g}^f.$$

Therefore

$$\dim \mathfrak{g}^e = \dim(\mathfrak{g}^f)^*.$$

Since φ is an injective linear map between finite-dimensional vector spaces of the same dimension, it is an isomorphism. Finally, we obtain

$$\kappa\big(f + \mathfrak{g}^e\big) = \chi + (\mathfrak{g}^f)^*.$$

$\square$

We now introduce a generalization of $\mathfrak{m}$. Let $\mathfrak{L}$ be an isotropic subspace of $\mathfrak{g}_{\frac{1}{2}}$, that is, $\omega_\chi(\mathfrak{L}, \mathfrak{L}) = 0$, and define

$$\mathfrak{m}_{\chi,\mathcal{L}} := \mathfrak{L} \oplus \bigoplus_{j>\frac{1}{2}} \mathfrak{g}_j, \qquad \mathfrak{n}_{\chi,\mathcal{L}} := \mathcal{L}^{\perp_{\omega_\chi}} \oplus \bigoplus_{j>\frac{1}{2}} \mathfrak{g}_j,$$

where

$$\mathcal{L}^{\perp_{\omega_\chi}} = \left\{ x \in \mathfrak{g}_{\frac{1}{2}} \mid \omega_\chi(x, \mathcal{L}) = 0 \right\}.$$

Notice that $\mathfrak{g}^e \subset \mathfrak{m}^\perp$, because $\mathfrak{g}^e$ is spanned by highest weight vectors. Here $\perp$ denotes the orthogonal complement with respect to the bilinear form $(\ |\ )$.

Let $\mathbf{N}$ be the unipotent subgroup of G whose Lie algebra is $\mathfrak{n}$. Thus it is connected [Hum75, Exercise 15.6]. When $\mathfrak{m} = \mathfrak{n}$, the subspace $\mathfrak{L}$ is Lagrangian, because for every symplectic space (V, ω) one has $\dim L + \dim L^{\perp\omega} = \dim V$. We also denote by $\mathbf{M}$ the closed connected subgroup of G whose Lie algebra is $\mathfrak{m}$ [Hum75, § 13.1].

Consider the adjoint map

$$\mathbf{N} \times (f + \mathfrak{m}^\perp) \longrightarrow \mathfrak{g}, \quad (g, x) \mapsto g \cdot x.$$

Proposition 3.20. *Its image is contained in* $f + \mathfrak{m}^\perp$.

Proof. For any $x \in \mathfrak{n}$ and any $y \in \mathfrak{m}^\perp$, we have $\exp(\operatorname{ad} x)(f + y) \in f + \mathfrak{m}^\perp$, because $[\mathfrak{n}, \mathfrak{m}] \subset \mathfrak{m}$ and $\chi([\mathfrak{n}, \mathfrak{m}]) = 0$. Therefore, for any $m \in \mathfrak{m}$,

$$\begin{aligned}\big(\exp(\operatorname{ad} x)(f + y) - f \big| m\big) &= -(f \big| [f, m]) + \big(\exp(\operatorname{ad} x)(f + y) \big| m\big) \\ &= 0 + \big(\exp(\operatorname{ad} x)(f) \big| m\big) + \big(\exp(\operatorname{ad} y) \big| m\big),\end{aligned}$$

and since $\mathfrak{n}$ is ad-nilpotent,

$$\begin{aligned}&= \big(f \big| \exp(\operatorname{ad}(-x))(m)\big) + 0 \\ &= 0.\end{aligned}$$

This is enough to prove the claim, because $\mathbf{N}$ is generated by all $\exp(\operatorname{ad} x)$ with x ranging over $\mathfrak{n}$. Hence, by restriction, we obtain a map

$$\alpha\colon \mathbf{N} \times \mathcal{S}_f \longrightarrow f + \mathfrak{m}^{\perp}.$$

□

By [GG02, § 2.3], we have the following theorem.

Theorem 3.21. *The map α is an isomorphism of affine algebraic varieties.*

Therefore the induced pullback

$$\alpha^*\colon \mathbb{C}[f + \mathfrak{m}^{\perp}] \longrightarrow \mathbb{C}[\mathbf{N} \times \mathcal{S}_f]$$

is an isomorphism. Consider the $\mathbf{N}$-equivariant actions on $\mathbf{N} \times \mathcal{S}_f$ and on $f + \mathfrak{m}^{\perp}$ with respect to α. The action on $\mathbf{N} \times \mathcal{S}_f$ is given by left translation on the first factor, namely

$$n \cdot (g, s) := (ng, s), \qquad n, g \in \mathbf{N},\ s \in \mathcal{S}_f,$$

while the action on $f + \mathfrak{m}^{\perp}$ is the adjoint action, that is,

$$n \cdot (g \cdot s) = (ng) \cdot s \qquad \text{for all } n, g \in \mathbf{N},\ s \in \mathcal{S}_f.$$

Now, since α is $\mathbf{N}$-equivariant, we have

$$\mathbb{C}[f + \mathfrak{m}^{\perp}]^{\mathbf{N}} \cong \mathbb{C}[\mathbf{N} \times \mathcal{S}_f]^{\mathbf{N}} \cong \mathbb{C}[\mathbf{N}]^{\mathbf{N}} \otimes \mathbb{C}[\mathcal{S}_f],$$

and

$$\mathbb{C}[\mathbf{N}]^{\mathbf{N}} \cong \mathbb{C}.$$

Thus we obtain

$$\mathbb{C}[\mathcal{S}_f] \cong \mathbb{C}[f + \mathfrak{m}^{\perp}]^{\mathbf{N}}.$$

After this, we will identify $\mathfrak{m}^\perp$ with $\mathfrak{m}^\circ := \{\xi \in \mathfrak{g}^* \mid \xi(m) = 0\}$ by means of $(\,|\,)$.

We now explicitly describe the Poisson bracket on $\mathbb{C}[\mathfrak{g}^*]$ [CG10, Proposition 1.3.18]. Let $(x_1, \ldots, x_n)$ be a basis of $\mathfrak{g}$ such that $[x_i, x_j] = \sum_k c_{i,j}^k x_k$. By the canonical isomorphism $(\mathfrak{g}^*)^* \cong \mathfrak{g}$, every element of $\mathfrak{g}$ may be regarded as a linear function on $\mathfrak{g}^*$, and hence also as an element of $\mathbb{C}[\mathfrak{g}^*]$. For $f, g \in \mathbb{C}[\mathfrak{g}^*]$, we have

$$\{f, g\} = \sum_{i,j,k} c_{i,j}^k x_k \frac{\partial f}{\partial x_i^*} \frac{\partial g}{\partial x_j^*}.$$

More concisely, we have

$$\{f, g\} \colon \mathfrak{g}^* \to \mathbb{C}, \quad \xi \mapsto \langle \xi, [d_\xi f, d_\xi g] \rangle, \tag{3.ii}$$

where $d_\xi f, d_\xi g \in (\mathfrak{g}^*)^* \cong \mathfrak{g}$ denote the differentials of f and g at $\xi \in \mathfrak{g}^*$. In particular, if $x, y \in \mathfrak{g} \cong (\mathfrak{g}^*)^* \subset \mathbb{C}[\mathfrak{g}^*]$, then

$$\{x, y\} = [x, y]. \tag{3.iii}$$

Moreover, for any regular functions $f, g \in \mathbb{C}[\mathfrak{g}^*]$ and any coadjoint orbit $\mathbb{O} \subset \mathfrak{g}^*$, we have

$$\{f, g\}|_{\mathbb{O}} = \{f|_{\mathbb{O}}, g|_{\mathbb{O}}\}_{\text{symplectic}}.$$

The bracket on the right-hand side comes from the symplectic structure on $\mathbb{O}$, while the bracket on the left-hand side comes from the restriction of the Poisson bracket on $\mathbb{C}[\mathfrak{g}^*]$ to $\mathbb{O}$ [CG10, Proposition 1.3.21].

Definition 3.22. If X is smooth, then it can be regarded as a complex-analytic manifold with a holomorphic Poisson structure. For each point $x \in X$, the *symplectic leaf* $\mathcal{S}_x$ through x is defined to be the set of points which can be reached from x by flowing along Hamiltonian vector fields.

Example 3.23. The space $\mathfrak{g}^*$ is a smooth Poisson variety, and its Hamiltonian vector fields are given by all $\{f, \cdot\}$ with $f \in \mathbb{C}[\mathfrak{g}^*]$. Moreover, the symplectic leaves of $\mathfrak{g}^*$ are precisely the coadjoint orbits in $\mathfrak{g}^*$ [Vai94, proposition 3.1].

By [Wei83], the natural Kirillov–Kostant–Souriau Poisson structure on $\mathfrak{g}^*$ can be induced on $\mathcal{S}_f$. Namely, the bracket $\{\ ,\ \}_{\mathcal{S}_f}$ on $\mathbb{C}[\mathcal{S}_f]$ is given by

$$\{f, g\}_{\mathcal{S}_f}(\xi) = \{f|_{\mathbb{O}}, g|_{\mathbb{O}}\}_{\text{symplectic}}(\xi), \tag{3.iv}$$

where, for any $f, g \in \mathbb{C}[\mathcal{S}_f]$ and $\xi \in \mathcal{S}_f$, $\mathbb{O}$ denotes the coadjoint orbit in $\mathfrak{g}^*$ through ξ.

However, for the convenience of quantization, we introduce another way to obtain the Poisson structure, namely by Hamiltonian reduction.

Definition 3.24. An action of a Lie algebra $\mathfrak{a}$ on a manifold X is called *Hamiltonian* if there exists a Lie algebra homomorphism

$$\tilde{\mu}\colon \mathfrak{a} \longrightarrow \mathcal{O}_X(X), \qquad a \longmapsto \tilde{\mu}_a,$$

such that the following diagram commutes:

$$\begin{tikzcd} \mathfrak{a} \arrow[r] \arrow[dr, dashed, "\tilde{\mu}"'] & \mathfrak{X}(X) \\ & \mathcal{O}_X(X) \arrow[u] \end{tikzcd}$$

Here $\mathfrak{X}(X)$ denotes the Lie algebra of symplectic vector fields on X, the vertical arrow is the natural embedding $\mathcal{O}_X(X) \hookrightarrow \mathfrak{X}(X)$ given by $f \mapsto \xi_f = \{f, \cdot\}$, which is a derivation on $\mathcal{O}_X(X)$, and the horizontal map $\mathfrak{a} \to \mathfrak{X}(X)$ is induced by the action of $\mathfrak{a}$ on X, explicitly:

$$a \longmapsto (x \longmapsto \text{the infinitesimal action in the direction } a \text{ at } x \in T_xX)\,.$$

We call $\tilde{\mu}$ the *co-moment map* of the action, or the *Hamiltonian* of the action. Its dual map

$$\mu\colon X \longrightarrow \mathfrak{a}^*, \qquad x \longmapsto \mu(x),$$

defined by requiring that $\langle \mu(x), a\rangle = \tilde{\mu}_a(x)$ for all $a \in \mathfrak{a}$, is called the *moment map* of the action.

Remark 3.25. If the group $\mathbf{A}$ is connected, then μ is $\mathbf{A}$-equivariant with respect to the coadjoint action on $\mathfrak{a}^*$.

Definition 3.26. Let X be a Poisson variety, let $Y \subset X$ be a subvariety, let $\iota\colon Y \hookrightarrow X$ be the inclusion map, and let $\pi\colon Y \twoheadrightarrow Z$ be a surjective morphism. Suppose that there exists a Poisson structure $\{\ ,\ \}_Z$ on Z such that for any open set $U \subset X$ and any

$$f, g \in \mathcal{O}_Z\big(\pi(U \cap Y)\big),$$

one has

$$\{f, g\}_Z \circ \pi(u) = \{\tilde{f}, \tilde{g}\} \circ \iota(u), \qquad \forall\, u \in U \cap Y,$$

where $\tilde{f}, \tilde{g} \in \mathcal{O}_X(U)$ are arbitrary extensions of

$$f \circ \pi|_{U \cap Y}, \qquad g \circ \pi|_{U \cap Y}$$

to U. Then the triple (X, Y, Z) is called Poisson reducible.

The following theorem plays a central role in Hamiltonian reduction [LPV13, proposition 5.39]:

Theorem 3.27 (Marsden–Weinstein)**.** *Let $\mathbf{A}$ be a connected algebraic group, let X be a Poisson variety, and suppose that the action of $\mathbf{A}$ on X is Hamiltonian. Let*

$$\mu\colon X \to \mathfrak{a}^*$$

be the moment map, and let $\zeta \in \mathfrak{a}^*$ be a regular value of μ. If the quotient

$$\mu^{-1}(\zeta)/\mathbf{A}$$

exists and is an algebraic variety, then $\mu^{-1}(\zeta)$ is an $\mathbf{A}$-stable subvariety. Denote by

$$\iota\colon \mu^{-1}(\zeta) \hookrightarrow X, \qquad \pi\colon \mu^{-1}(\zeta) \twoheadrightarrow \mu^{-1}(\zeta)/\mathbf{A}$$

the inclusion map and the quotient map respectively. Then the triple

$$\big(X, \mu^{-1}(\zeta), \mu^{-1}(\zeta)/\mathbf{A}\big)$$

is Poisson reducible. In particular, $\mu^{-1}(\zeta)/\mathbf{A}$ *naturally inherits a Poisson structure.*

Remark 3.28. The Poisson structure obtained in this theorem is regarded as being obtained by Hamiltonian reduction.

Assume $\mathfrak{m} = \mathfrak{n}$, and in this case let $\mathbf{M}$ be the unipotent subgroup of $\mathbf{G}$ whose Lie algebra is $\mathfrak{m}$.

We intend to apply this theorem to the connected Lie group $\mathbf{M}$ acting on the algebraic variety $\mathfrak{g}^*$, where the algebra $\mathcal{O}_{\mathfrak{g}^*}(\mathfrak{g}^*)$ is equipped with the Lie–Poisson bracket.

Proposition 3.29. *This action is Hamiltonian, and the moment map*

$$\mu\colon \mathfrak{g}^* \to \mathfrak{m}^*$$

is simply the restriction of a linear functional on $\mathfrak{g}$ *to* $\mathfrak{m}$.

Proof. For a regular function $f : \mathfrak{g}^* \to \mathbb{C}$, its differential at $\xi \in \mathfrak{g}^*$ is a linear function

$$df_\xi : T_\xi(\mathfrak{g}^*) \to \mathbb{C}, \qquad df_\xi(v) = \frac{d}{dt}\bigg|_{t=0} f(\xi + tv).$$

Since $\mathfrak{g}^*$ is a vector space, we have an isomorphism

$$T_\xi(\mathfrak{g}^*) \cong \mathfrak{g}^*,$$

and hence

$$df_\xi \in (\mathfrak{g}^*)^* \cong \mathfrak{g}.$$

In particular, for the linear function $f_x(\xi) := \xi(x)$, where $x \in \mathfrak{g}$, we have

$$df_{x,\xi} = x.$$

Consider the coadjoint action of M on $\mathfrak{g}^*$. Define

$$\tilde{\mu} : \mathfrak{m} \to \mathcal{O}_{\mathfrak{g}^*}(\mathfrak{g}^*), \qquad \tilde{\mu}_x(\xi) := \xi(x), \quad x \in \mathfrak{m},\ \xi \in \mathfrak{g}^*.$$

This means that $d\tilde{\mu}_{x,\xi} = x$. Then

$$\tilde{\mu} : \mathfrak{m} \to \mathcal{O}_{\mathfrak{g}^*}(\mathfrak{g}^*), \qquad x \mapsto \tilde{\mu}_x$$

is the co-moment map. Indeed, by 3.ii, for $x, y \in \mathfrak{m}$,

$$\{\tilde{\mu}_x, \tilde{\mu}_y\}(\xi) = \xi([x, y]) = \tilde{\mu}_{[x,y]}(\xi),$$

so $\tilde{\mu}$ is a Lie algebra homomorphism. By definition, the corresponding moment map

$$\mu : \mathfrak{g}^* \to \mathfrak{m}^*$$

is the dual map characterized by

$$\mu(\xi)(x) = \tilde{\mu}_x(\xi) = \xi(x), \qquad x \in \mathfrak{m}.$$

Therefore

$$\mu(\xi) = \xi|_{\mathfrak{m}}.$$

We now show that the coadjoint action of M on $\mathfrak{g}^*$ is Hamiltonian. It remains to verify that $\tilde{\mu}_x$ generates the infinitesimal coadjoint action induced by x. For $y \in \mathfrak{g} \cong (\mathfrak{g}^*)^*$, we do not distinguish between y and f_y. By 3.iii, we have

$$\begin{aligned}\left(\frac{d}{dt}\bigg|_{t=0} \exp(t\,\mathrm{ad}_x^*)\xi\right)(y) &= \frac{d}{dt}\bigg|_{t=0} \big((\exp(t\,\mathrm{ad}_x^*)\xi)(y)\big) \\ &= (\mathrm{ad}_x^*\,\xi)(y) \\ &= -\xi([x, y]).\end{aligned}$$

On the other hand,

$$\{\tilde{\mu}_x, f_y\}(\xi) = \xi([x, y]).$$

Therefore the Hamiltonian vector field of $\tilde{\mu}_x$ agrees with the infinitesimal vector field induced by x. Hence the coadjoint action of M on $\mathfrak{g}^*$ is Hamiltonian. $\square$

Since $\chi|_{\mathfrak{m}}$ is a character of $\mathfrak{r}$, we have

$$\big(\operatorname{ad}^* m(\chi)\big)y = -\chi([x,y]) = 0 \quad x, y \in \mathfrak{m}.$$

Therefore

$$\operatorname{ad}^* x(\chi|_{\mathfrak{m}}) = 0.$$

That is,

$$\operatorname{Ad}^* m(\chi|_{\mathfrak{m}}) = \chi|_{\mathfrak{m}} \quad m \in M.$$

Hence the set

$$\mu^{-1}(\chi|_{\mathfrak{m}}) = \{\xi \in \mathfrak{g}^* \mid \mu(\xi) = \chi|_{\mathfrak{m}}\}$$

is **M**-stable. Moreover,

$$\mu(\xi) = \chi|_{\mathfrak{m}} \iff (\xi - \chi)|_{\mathfrak{m}} = 0 \iff \xi - \chi \in \mathfrak{m}^\circ \cong \mathfrak{m}^\perp.$$

Therefore

$$\mu^{-1}(\chi|_{\mathfrak{m}}) = \chi + \mathfrak{m}^\perp.$$

In addition, we have the following lemma.

Lemma 3.30. *[AM16, Lemma 3.23] For every symplectic leaf of $\mathfrak{g}^*$, $\chi|_{\mathfrak{m}}$ is a regular value of the restriction of μ.*

Since the map

$$\mathbf{M} \times \mathcal{S}_f \longrightarrow \chi + \mathfrak{m}^\perp$$

is an isomorphism of affine algebraic varieties, we have

$$\mathcal{S}_f \cong (\chi + \mathfrak{m}^\perp)/\mathbf{M} = \mu^{-1}(\chi|_{\mathfrak{m}})/\mathbf{M}.$$

Thus the conditions of Theorem 3.27 are satisfied. We obtain a Poisson structure on $\mathcal{S}_f$. It is described as follows.

Let $\pi\colon \chi + \mathfrak{m}^\perp \twoheadrightarrow (\chi + \mathfrak{m}^\perp)/\mathbf{M} \simeq \mathcal{S}_f$ be the natural projection map, and let $\iota\colon \chi + \mathfrak{m}^\perp \hookrightarrow \mathfrak{g}^*$ be the natural inclusion map. Then, for any $f, g \in \mathbb{C}[\mathcal{S}_f]$,

$$\{f, g\}_{\mathcal{S}_f} \circ \pi = \{\tilde{f}, \tilde{g}\} \circ \iota,$$

where $\tilde{f}, \tilde{g}$ are arbitrary extensions of $f \circ \pi, g \circ \pi$ to $\mathfrak{g}^*$. Thus the Poisson structure defined on $\mathcal{S}_f$ in this way is the same as the one defined in 3.iv.

Let

$$I_\chi = \mathbb{C}[\mathfrak{g}^*] \sum_{x \in \mathfrak{m}} (x - \chi(x)),$$

here $\mathbb{C}[\mathfrak{g}^*]$ is identified with $\mathbb{C}[\mathfrak{g}]$, and hence

$$\mathbb{C}[\mu^{-1}(\chi)] = \mathbb{C}[\mathfrak{g}^*]/I_\chi.$$

Then $\mathbb{C}[\mathcal{S}_f] = \mathbb{C}[\mu^{-1}(\chi)]^{\mathbf{M}}$ can be identified with the subspace of $\mathbb{C}[\mathfrak{g}^*]/I_\chi$ consisting of all cosets $\phi + I_\chi$ such that $\{x, \phi\} \in I_\chi$ for every $x \in \mathfrak{m}$. Under this realization, the Poisson structure on $\mathbb{C}[\mathcal{S}_f]$ is defined by the formula

$$\{\phi + I_\chi, \phi' + I_\chi\} = \{\phi, \phi'\} + I_\chi,$$

where ϕ, ϕ' satisfy $\{x, \phi\}, \{x, \phi'\} \in I_\chi$ for every $x \in \mathfrak{m}$.

Example 3.31. Write a general element of $\mathfrak{sl}_4$ as

$$X = (x_{ij})_{1 \le i,j \le 4}, \qquad \sum_{i=1}^{4} x_{ii} = 0.$$

Since

$$\mathfrak{m} = \operatorname{span}\{E_{12}, E_{14}, E_{23}, E_{43}, E_{13}\},$$

and $\chi + \mathfrak{m}^\perp \in \mathfrak{g}^*$ is isomorphic to $f + \mathfrak{m}^\perp$, a general point of $\chi + m^\perp$ has the form

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ 2 & x_{22} & x_{23} & x_{24} \\ 0 & 2 & x_{33} & 0 \\ 0 & x_{42} & x_{43} & x_{44} \end{pmatrix}, \qquad x_{44} = -(x_{11} + x_{22} + x_{33}).$$

Unfortunately, I am unable to compute an explicit expression for $\mathbb{C}[\mu^{-1}(\chi)]^{\mathbf{M}}$ in the $\mathfrak{sl}_4$ case, or in any nontrivial case. Next we give a brief explanation of why this computation is very difficult. Recall that $\alpha : M \times S_f \longrightarrow f + m^{\perp}$ is an isomorphism of affine algebraic varieties. Recalling the whole process, we have

$$(\mathbb{C}[\mathfrak{g}^*]/I_\chi)^M \longrightarrow \mathbb{C}^M[\mu^{-1}(\chi|_m)] \longrightarrow \mathbb{C}^M[f+m^{\perp}] \longrightarrow \mathbb{C}^M[M\times S_f] \longrightarrow \mathbb{C}[S_f].$$

We may choose suitable representatives G, so that on elements the above maps can be written as

$$\overline{G} \longmapsto G \longmapsto G\circ k \longmapsto G\circ k\circ\alpha \longmapsto \sum (G\circ k\circ\alpha)_1 \otimes (G\circ k\circ\alpha)_2,$$

where

$$\begin{aligned} \kappa :\ f + m^{\perp} &\longrightarrow \chi + m^{\circ} \\ F &\longmapsto (F|\ ), \end{aligned}$$

and the map between $\mathbb{C}^M[M \times S_f]$ and $\mathbb{C}[S_f]$ is given on elements as follows:

$$\begin{aligned} \mathbb{C}^M[M \times S_f] &\xrightarrow{\sim} \mathbb{C}^M[M] \otimes \mathbb{C}[S_f] \\ F &\longmapsto \sum F_1 \otimes F_2 \quad \text{s.t. } F(m,s) = \sum F_1(m)F_2(s). \end{aligned}$$

It is easy to see that it is difficult to describe explicit elements of $\mathbb{C}^M[M\times S_f]$, and that it is extremely difficult to give a general explicit description of the map from this algebra to $\mathbb{C}[S_f]$. Since all the arrows above are isomorphisms, it is natural to consider the arrows in the reverse direction. We next explain that the difficulty still cannot be avoided.

$$\mathbb{C}[S_f] \longrightarrow \mathbb{C}^M[M\times S_f] \longrightarrow \mathbb{C}^M[f+m^{\perp}] \longrightarrow \mathbb{C}^M[\mu^{-1}(\chi|_{\mathfrak{m}})] \longrightarrow (\mathbb{C}[\mathfrak{g}^*]/I_\chi)^M.$$

On elements, we have

$$G \longmapsto (1,G) \longmapsto (1,G)\cdot\alpha^{-1} \longmapsto (1,G)\cdot\alpha^{-1}\cdot k^{-1} \longmapsto \overline{(1,G)\cdot\alpha^{-1}\cdot k^{-1}},$$

where

$$\kappa^{-1}:\ \chi + m^{\circ} \longrightarrow f + m^{\perp}$$

$$H(\cdot) = (f_H|\cdot) \longmapsto f_H.$$

It is easy to see that, for an arbitrary element of $\mu^{-1}(\chi|_{\mathfrak{m}})$, its image under α^{-1} is almost impossible to find. Moreover, the image of κ^{-1} is also difficult to describe. Precisely for this reason, in concrete examples, we cannot obtain this Poisson structure by generators and relations. Nevertheless, we can still continue to state the quantization of Slodowy slices.

Chapter 4

Quantization of Slodowy Slices and Finite W-Algebras

In this chapter, we discuss BRST cohomology and the quantization of Slodowy slices. In Section 4.1, we define BRST cohomology using a special graded Clifford algebra, and ultimately show that this cohomology realizes the Poisson structure on the Slodowy slice. In Section 4.2, we introduce the quantum BRST cohomology and explain that it gives a quantization of the Slodowy slice. In Section 4.3, we give the classical definition of finite W-algebras and show that it coincides with the quantization of the Slodowy slice. The material in this chapter is discussed in detail in [Ara17], and we follow the notation used there.

4.1 The BRST realization of Slodowy slices

This part is explained in detail in [Ara17, § 2]. We follow his notation.

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra, let G be its connected affine algebraic group, let $\{x_i\}_{1\leq i\leq d}$ be a basis of $\mathfrak{g}$, and let $\{x_i^*\}_{1\leq i\leq d}$ be the corresponding dual

basis of $\mathfrak{g}^*$. For $i, j = 1, \ldots, d$, write

$$[x_i, x_j] = \sum_{k=1}^{d} c_{i,j}^k x_k.$$

Let $Cl(\mathfrak{g})$ be the *Clifford algebra* associated with the finite-dimensional vector space $\mathfrak{g} \oplus \mathfrak{g}^*$ and the non-degenerate bilinear form

$$B(\cdot, \cdot) := \langle \cdot | \cdot \rangle,$$

where

$$\langle f + x \mid g + y \rangle = f(y) + g(x)$$

for $f, g \in \mathfrak{g}^*$ and $x, y \in \mathfrak{g}$. Hence its Lie superbracket is given as follows. For homogeneous elements $a, b \in Cl(\mathfrak{g})$,

$$[a, b] = ab - (-1)^{|a||b|} ba.$$

A direct computation shows that for homogeneous elements $a, b, c \in Cl(\mathfrak{g})$, the Leibniz rule holds:

$$[a, bc] = [a, b]c + (-1)^{|a||b|} b[a, c].$$

In particular,

$$[x, f] = f(x) \quad x \in \mathfrak{g} \subset \wedge\mathfrak{g},\ f \in \mathfrak{g}^* \subset \wedge\mathfrak{g}^*.$$

The algebra $Cl(\mathfrak{g})$ is a unital superalgebra. By 2.i and [Mei13, § 2.1.2], as a $\mathbb{C}$-linear space it is isomorphic to

$$\wedge(\mathfrak{g} \oplus \mathfrak{g}^*) \cong \wedge(\mathfrak{g}) \otimes \wedge(\mathfrak{g}^*).$$

Moreover, the natural embeddings

$$\wedge(\mathfrak{g}) \hookrightarrow Cl(\mathfrak{g}), \qquad \wedge(\mathfrak{g}^*) \hookrightarrow Cl(\mathfrak{g})$$

are homogeneous homomorphisms of superalgebras.

Define an increasing filtration on $Cl(\mathfrak{g})$ by

$$Cl_p(\mathfrak{g}) := \wedge^{\leqslant p}(\mathfrak{g}) \otimes \wedge(\mathfrak{g}^*).$$

We have

$$0 = Cl_{-1}(\mathfrak{g}) \subset Cl_0(\mathfrak{g}) \subset Cl_1(\mathfrak{g}) \subset \cdots \subset Cl_d(\mathfrak{g}) = Cl(\mathfrak{g}),$$

and

$$Cl_p(\mathfrak{g}) \cdot Cl_q(\mathfrak{g}) \subset Cl_{p+q}(\mathfrak{g}), \qquad [Cl_p(\mathfrak{g}), Cl_q(\mathfrak{g})] \subset Cl_{p+q-1}(\mathfrak{g}).$$

The first formula is easy to prove. To verify the second one, take

$$x \in Cl_p(\mathfrak{g}) \setminus Cl_{p-1}(\mathfrak{g}), \qquad y \in Cl_q(\mathfrak{g}) \setminus Cl_{q-1}(\mathfrak{g}).$$

It is enough to check that $[x, y] \in Cl_{q+p-1}(\mathfrak{g})$, and this follows directly from the definition.

Let $\overline{Cl}(\mathfrak{g})$ be the associated graded algebra:

$$\overline{Cl}(\mathfrak{g}) := \operatorname{gr} Cl(\mathfrak{g}) = \bigoplus_{p \geqslant 0} \frac{Cl_p(\mathfrak{g})}{Cl_{p-1}(\mathfrak{g})}.$$

The algebra $\overline{Cl}(\mathfrak{g})$ has a graded Poisson superalgebra structure related to the Lie superbracket. It is called the *classical Clifford algebra* associated with $\mathfrak{g} \oplus \mathfrak{g}^*$. Its Poisson superbracket is given by

$$\{x, f\} = f(x), \qquad x \in \mathfrak{g} \subset \wedge(\mathfrak{g}), \quad f \in \mathfrak{g}^* \subset \wedge(\mathfrak{g}^*),$$

$$\{x, y\} = 0, \qquad x, y \in \mathfrak{g} \subset \wedge(\mathfrak{g}),$$

$$\{f, g\} = 0, \qquad f, g \in \mathfrak{g}^* \subset \wedge(\mathfrak{g}^*).$$

Remark 4.1. These formulas are enough to define the whole Poisson superalgebra structure. We give an example to illustrate the computation.

Example 4.2. Take

$$A = x_a x_b x_c^* \in Cl_2(\mathfrak{g}), \qquad B = x_d x_e^* \in Cl_1(\mathfrak{g}).$$

Using the above relations and 2.26, let us compute $\{A, B\}$. Since A is odd, the super Leibniz rule gives

$$\{x_a x_b x_c^*, x_d x_e^*\} = \{x_a x_b x_c^*, x_d\} x_e^* - x_d \{x_a x_b x_c^*, x_e^*\}.$$

Here

$$\begin{aligned}\{x_a x_b x_c^*, x_d\} &= x_a x_b \{x_c^*, x_d\} - \{x_a x_b, x_d\} x_c^* \\ &= \delta_{cd} x_a x_b.\end{aligned}$$

Moreover,

$$\begin{aligned}\{x_a x_b x_c^*, x_e^*\} &= x_a x_b \{x_c^*, x_e^*\} - \{x_a x_b, x_e^*\} x_c^* \\ &= -\delta_{be} x_a x_c^* + \delta_{ae} x_b x_c^*.\end{aligned}$$

Substituting these into the original formula, we get

$$\begin{aligned}\{x_a x_b x_c^*, x_d x_e^*\} &= \delta_{cd} x_a x_b x_e^* - x_d \left(-\delta_{be} x_a x_c^* + \delta_{ae} x_b x_c^*\right) \\ &= \delta_{cd} x_a x_b x_e^* + \delta_{be} x_d x_a x_c^* - \delta_{ae} x_d x_b x_c^* \\ &= \delta_{cd} x_a x_b x_e^* - \delta_{be} x_a x_d x_c^* + \delta_{ae} x_b x_d x_c^*.\end{aligned}$$

In particular, it is easy to see that

$$\{x_a x_b x_c^*, x_d x_e^*\} \in Cl_2(\mathfrak{g}).$$

This agrees with

$$\{Cl_p(\mathfrak{g}), Cl_q(\mathfrak{g})\} \subset Cl_{p+q-1}(\mathfrak{g})$$

in the case $p = 2, q = 1$.

Let

$$\bar{C}(\mathfrak{g}) = \mathbb{C}[\mathfrak{g}^*] \otimes \overline{Cl}(\mathfrak{g}).$$

Then $\bar{C}(\mathfrak{g})$ is naturally a Poisson superalgebra, because it is the tensor product of Poisson superalgebras, where the superalgebra structure on $\mathbb{C}[\mathfrak{g}^*]$ is given by its grading modulo 2 as a polynomial ring.

A direct verification gives the following result [AM24, lemma 7.1]:

Lemma 4.3. *The following map is a Lie algebra homomorphism:*

$$\rho : \mathfrak{g} \to Cl(\mathfrak{g}), \quad x_i \mapsto \sum_{1 \leq j,k \leq d} c_{i,j}^k x_k x_j^*,$$

and for $x, y \in \mathfrak{g}$, one has

$$[\rho(x), y] = [x, y] \in \mathfrak{g} \subset Cl(\mathfrak{g}),$$

where the first bracket is the Lie bracket in $Cl(\mathfrak{g})$ and the second is the Lie bracket in $\mathfrak{g}$.

Remark 4.4. The map $\rho : \mathfrak{g} \to Cl_1(\mathfrak{g}) \subset Cl(\mathfrak{g})$ naturally induces a Lie algebra homomorphism

$$\bar{\rho} := \sigma_1 \circ \rho : \mathfrak{g} \to \bar{Cl}(\mathfrak{g}),$$

where σ_1 is the projection

$$Cl_1(\mathfrak{g}) \to Cl_1(\mathfrak{g})/Cl_0(\mathfrak{g}) \subset \operatorname{gr} Cl(\mathfrak{g}).$$

Lemma 4.5. *The following map is a Lie algebra homomorphism:*

$$\begin{aligned} \bar{\theta}_\chi : \mathfrak{g} &\longrightarrow \bar{C}(\mathfrak{g}) \\ x &\longmapsto (x - \chi(x)) \otimes 1 + 1 \otimes \bar{\rho}(x). \end{aligned}$$

Proof. This follows by a direct computation, using the fact that χ is a character on $\mathfrak{g}$. □

Let

$$\bar{C}(\mathfrak{g}) = \bigoplus_{n\in\mathbb{Z}} \bar{C}^n(\mathfrak{g})$$

be the $\mathbb{Z}$-grading defined by

$$\bar{C}^n(\mathfrak{g}) = \mathbb{C}[\mathfrak{g}^*] \otimes \left(\bigoplus_{j-i=n} \wedge^i(\mathfrak{g}) \otimes \wedge^j(\mathfrak{g}^*) \right).$$

Here

$$\deg(\phi \otimes 1) = 0 \quad \phi \in \mathbb{C}[\mathfrak{g}^*],$$

$$\deg(1 \otimes f) = 1 \quad f \in \mathfrak{g}^*,$$

$$\deg(1 \otimes x) = -1 \quad x \in \mathfrak{g}.$$

Lemma 4.6. *There exists a unique element $\bar{D} \in \bar{C}^1(\mathfrak{g})$ such that*

$$\{\bar{D}, 1 \otimes x\} = \bar{\theta}_\chi(x) \qquad \textit{for } x \in \mathfrak{g}.$$

Moreover,

$$\{\bar{D}, \bar{D}\} = 0.$$

Proof. First notice that

$$\overline{Cl}(\mathfrak{g})^{\mathfrak{g}} := \{w \in \overline{Cl}(\mathfrak{g}) \mid \{x, w\} = 0,\ \forall x \in \mathfrak{g}\} = \wedge(\mathfrak{g}).$$

Let

$$\bar{D} = \sum_i \big(x_i - \chi(x_i)\big) \otimes x_i^* - 1 \otimes \frac{1}{2} \sum_{i,j,k} c_{i,j}^k x_i^* x_j^* x_k.$$

Using the super Leibniz rule and

$$\{x_i^*, x_j\} = \delta_{ij}, \qquad \{x_i, x_j\} = 0,$$

a direct computation gives

$$\{\bar{D}, 1 \otimes x\} = (x - \chi(x)) \otimes 1 + 1 \otimes \bar{\rho}(x) = \bar{\theta}_\chi(x).$$

This proves existence.

We prove uniqueness. Suppose that $\bar{D}_1, \bar{D}_2 \in \overline{C}^1(\mathfrak{g})$ both satisfy the property in the lemma. Let

$$R = \bar{D}_1 - \bar{D}_2.$$

Then

$$\{R, 1 \otimes x\} = 0, \qquad \forall x \in \mathfrak{g}.$$

Hence

$$R \in \mathbb{C}[\mathfrak{g}^*] \otimes \overline{Cl}(\mathfrak{g})^{\mathfrak{g}} = \mathbb{C}[\mathfrak{g}^*] \otimes \wedge(\mathfrak{g}).$$

But by the definition of the degree, all elements in

$$\mathbb{C}[\mathfrak{g}^*] \otimes \wedge(\mathfrak{g})$$

have degree at most 0. Therefore

$$\big(\mathbb{C}[\mathfrak{g}^*] \otimes \wedge(\mathfrak{g})\big) \cap \overline{C}^1(\mathfrak{g}) = 0.$$

Thus $R = 0$, and uniqueness is proved.

Finally, by the super Jacobi identity and

$$\{\bar{D}, 1 \otimes x\} = \bar{\theta}_\chi(x),$$
$$[\overline{\rho}(x), y] = [x, y],$$

a direct computation gives, for any $x, y \in \mathfrak{g}$,

$$\{1 \otimes x, \{1 \otimes y, \{\bar{D}, \bar{D}\}\}\} = 0.$$

Together with the above equality

$$\overline{Cl}(\mathfrak{g})^{\mathfrak{g}} = \wedge(\mathfrak{g}),$$

and by considering the degree, we obtain

$$\{\bar{D}, \bar{D}\} = 0.$$

□

Consider

$$\operatorname{ad} \bar{D} := \{\bar{D}, \cdot\}.$$

By Lemma 4.6 and the super Jacobi identity, it satisfies

$$(\operatorname{ad} \bar{D})^2 = 0.$$

Therefore

$$(\overline{C}(\mathfrak{g}), \operatorname{ad} \bar{D})$$

is a *differential graded Poisson superalgebra*. Its cohomology

$$H^{\bullet}_{BRST,\chi}(\mathfrak{g}, \mathbb{C}[\mathfrak{g}^*]) := H^{\bullet}(\overline{C}(\mathfrak{g}), \operatorname{ad} \bar{D}) = \bigoplus_{i \in \mathbb{Z}} H^i(\overline{C}(\mathfrak{g}), \operatorname{ad} \bar{D})$$

inherits a graded Poisson superalgebra structure from $\overline{C}(\mathfrak{g})$.

More generally, let P be a Poisson algebra equipped with a Poisson algebra homomorphism

$$\mu^* : \mathbb{C}[\mathfrak{g}^*] \to P.$$

Since the map

$$\mu^* \otimes \mathrm{id} : \overline{C}(\mathfrak{g}) \to P \otimes \overline{Cl}(\mathfrak{g})$$

is a Poisson superalgebra homomorphism, we may regard $\bar{D}$ as an element of

$$P \otimes \overline{Cl}(\mathfrak{g}).$$

Now

$$(P \otimes \overline{Cl}(\mathfrak{g}), \operatorname{ad} \bar{D})$$

is a differential graded Poisson superalgebra. Therefore its cohomology

$$H^{\bullet}_{BRST,\chi}(\mathfrak{g}, P) := H^{\bullet}(P \otimes \overline{Cl}(\mathfrak{g}), \operatorname{ad} \bar{D}) \tag{4.i}$$

inherits a graded Poisson superalgebra structure from

$$P \otimes \overline{Cl}(\mathfrak{g}).$$

Let $\mathfrak{m}$ be defined as in 3.i. Then we obtain the BRST realization of the Slodowy slice [Ara17, § 2.7]:

Theorem 4.7. *We have*

$$H^i_{BRST,\chi}(\mathfrak{m}, \mathbb{C}[\mathfrak{g}^*]) = 0 \qquad \textit{for } i \neq 0,$$

and

$$H^0_{BRST,\chi}(\mathfrak{m}, \mathbb{C}[\mathfrak{g}^*]) \cong \mathbb{C}[\mathcal{S}_f]$$

as Poisson algebras.

Next, we quantize Slodowy slices by quantum BRST cohomology.

4.2 Quantum BRST cohomology and quantization of Slodowy slices

The idea and the results of this section are completely parallel to those of the previous section. Let $\{U_i(\mathfrak{g})\}$ be the PBW filtration of the universal enveloping algebra $U(\mathfrak{g})$ of $\mathfrak{g}$. By the PBW theorem, there is an isomorphism of Poisson algebras

$$\operatorname{gr} U(\mathfrak{g}) = \bigoplus_{i \geq 0} U_i(\mathfrak{g})/U_{i-1}(\mathfrak{g}) \cong S(\mathfrak{g}) \cong \mathbb{C}[\mathfrak{g}^*].$$

Here the Poisson structure on $\mathbb{C}[\mathfrak{g}^*] \cong S(\mathfrak{g})$ is given by

$$\{x, y\} = [x, y], \qquad x, y \in \mathfrak{g} \subset S(\mathfrak{g}),$$

and is uniquely extended by the Leibniz rule.

Let

$$C(\mathfrak{g}) = U(\mathfrak{g}) \otimes Cl(\mathfrak{g}).$$

Here $U(\mathfrak{g})$ is regarded as a purely even superalgebra, so the $\mathbb{Z}_2$-grading of $C(\mathfrak{g})$ is completely determined by the $\mathbb{Z}_2$-grading of $Cl(\mathfrak{g})$. Using the filtrations on $U(\mathfrak{g})$ and $Cl(\mathfrak{g})$, define a filtration on $C(\mathfrak{g})$ by

$$C_p(\mathfrak{g}) = \sum_{i+j\leq p} U_i(\mathfrak{g}) \otimes Cl_j(\mathfrak{g}).$$

Recall that

$$\overline{C}(\mathfrak{g}) = \mathbb{C}[\mathfrak{g}^*] \otimes \overline{Cl}(\mathfrak{g}).$$

Then there is an isomorphism of graded Poisson superalgebras

$$\operatorname{gr} C(\mathfrak{g}) \cong \overline{C}(\mathfrak{g}).$$

Indeed, the map

$$\begin{aligned} \Phi :\ \operatorname{gr} U(\mathfrak{g}) \otimes \operatorname{gr} Cl(\mathfrak{g}) &\longrightarrow \operatorname{gr} C(\mathfrak{g}) \\ (u + U_{i-1}) \otimes (a + Cl_{j-1}) &\longmapsto u \otimes a + C_{i+j-1} \end{aligned}$$

is an isomorphism of Poisson superalgebras. Therefore $C(\mathfrak{g})$ can be regarded as a quantization of $\overline{C}(\mathfrak{g})$.

Define the following $\mathbb{Z}$-grading on $C(\mathfrak{g})$:

$$C^n(\mathfrak{g}) = U(\mathfrak{g}) \otimes \left(\bigoplus_{j-i=n} \wedge^i(\mathfrak{g}) \otimes \wedge^j(\mathfrak{g}^*) \right).$$

Here

$$\begin{aligned} \deg(u \otimes 1) &= 0, \qquad u \in U(\mathfrak{g}), \\ \deg(1 \otimes f) &= 1, \qquad f \in \mathfrak{g}^*, \\ \deg(1 \otimes x) &= -1, \qquad x \in \mathfrak{g}. \end{aligned}$$

Thus

$$C(\mathfrak{g}) = \bigoplus_{n\in\mathbb{Z}} C^n(\mathfrak{g}).$$

Recall the Lie algebra homomorphism defined earlier:

$$\rho : \mathfrak{g} \longrightarrow Cl(\mathfrak{g}), \qquad x_i \longmapsto \sum_{1\leq j,k\leq d} c_{i,j}^k x_k x_j^*.$$

Similarly, we have:

Lemma 4.8. *The map*

$$\theta_\chi : \mathfrak{g} \longrightarrow C(\mathfrak{g}), \qquad x \longmapsto (x - \chi(x)) \otimes 1 + 1 \otimes \rho(x)$$

is a Lie algebra homomorphism.

Proof. Since $\rho : \mathfrak{g} \to Cl(\mathfrak{g})$ is a Lie algebra homomorphism and $\chi([\mathfrak{g}, \mathfrak{g}]) = 0$, a direct computation gives

$$[\theta_\chi(x), \theta_\chi(y)] = \theta_\chi([x, y]), \qquad x, y \in \mathfrak{g}.$$

□

Lemma 4.9. *There exists a unique element $D \in C^1(\mathfrak{g})$ such that for every $x \in \mathfrak{g}$, one has*

$$[D, 1 \otimes x] = \theta_\chi(x).$$

Moreover,

$$D^2 = 0.$$

Proof. First notice that

$$Cl(\mathfrak{g})^{\mathfrak{g}} := \{w \in Cl(\mathfrak{g}) \mid [x, w] = 0,\ \forall x \in \mathfrak{g}\} = \wedge(\mathfrak{g}).$$

Let

$$D = \sum_i (x_i - \chi(x_i)) \otimes x_i^* - 1 \otimes \frac{1}{2} \sum_{i,j,k} c_{i,j}^k x_i^* x_j^* x_k.$$

Using the super Leibniz rule and

$$[x_i^*, x_j] = \delta_{ij}, \qquad [x_i, x_j] = 0,$$

a direct computation gives

$$[D, 1 \otimes x] = (x - \chi(x)) \otimes 1 + 1 \otimes \rho(x) = \theta_\chi(x).$$

This proves existence.

The proof of uniqueness and the proof of $D^2 = 0$ are the same as those in the previous section. $\square$

Consider

$$\operatorname{ad} D = [D, -].$$

By the above lemma and the super Jacobi identity, we know that

$$(\operatorname{ad} D)^2 = 0.$$

Therefore

$$(C(\mathfrak{g}), \operatorname{ad} D)$$

is a differential graded superalgebra. Its cohomology is denoted by

$$H^\bullet_{\mathrm{BRST},\chi}(\mathfrak{g}, U(\mathfrak{g})) := H^\bullet(C(\mathfrak{g}), \operatorname{ad} D) = \bigoplus_{i \in \mathbb{Z}} H^i(C(\mathfrak{g}), \operatorname{ad} D).$$

It inherits a graded superalgebra structure from $C(\mathfrak{g})$.

More generally, let A be a $U(\mathfrak{g})$-algebra. By the natural map

$$C(\mathfrak{g}) = U(\mathfrak{g}) \otimes Cl(\mathfrak{g}) \longrightarrow A \otimes Cl(\mathfrak{g}),$$

we still denote the image of D by D. Then

$$(A \otimes Cl(\mathfrak{g}), \operatorname{ad} D)$$

is a differential graded superalgebra. Therefore we define

$$H^\bullet_{\mathrm{BRST},\chi}(\mathfrak{g}, A) := H^\bullet(A \otimes Cl(\mathfrak{g}), \operatorname{ad} D).$$

When $\chi = 0$, we simply write

$$H^{\bullet}_{\mathrm{BRST}}(\mathfrak{g}, A).$$

Now take $\mathfrak{m}$ as defined above. Define

$$U(\mathfrak{g}, f) := H^{0}_{\mathrm{BRST},\chi}(\mathfrak{m}, U(\mathfrak{g})).$$

This algebra is called the finite W-algebra associated with f.

By the computation of classical BRST cohomology and the filtration isomorphism above, we have

$$\operatorname{gr} U(\mathfrak{g}, f) \cong \mathbb{C}[\mathcal{S}_f].$$

Therefore $U(\mathfrak{g}, f)$ is a quantization of the coordinate ring $\mathbb{C}[\mathcal{S}_f]$ of the Slodowy slice $\mathcal{S}_f$.

Finite W-algebras themselves are algebraic structures very worthy of study. See, for example, [BK06; Lu+25; AE22]. On the other hand, finite W-algebras also have a more algebraic and more commonly used definition, namely the definition by the Whittaker model.

4.3 The Whittaker model realization of finite W-algebras

Let

$$\chi = (f \mid \cdot) \in \mathfrak{g}^*.$$

By restriction, χ defines a one-dimensional representation

$$\chi : U(\mathfrak{m}) \to \mathbb{C}.$$

We denote the corresponding left $U(\mathfrak{m})$-module by $\mathbb{C}_\chi$.

Let I_χ be the left ideal of $U(\mathfrak{g})$ generated by the elements

$$x - \chi(x), \qquad x \in \mathfrak{m},$$

that is,

$$I_\chi := \sum_{x \in \mathfrak{m}} U(\mathfrak{g})(x - \chi(x)).$$

Then there is a natural isomorphism of left $U(\mathfrak{g})$-modules

$$Q_\chi := U(\mathfrak{g}) \otimes_{U(\mathfrak{m})} \mathbb{C}_\chi \cong U(\mathfrak{g})/I_\chi.$$

The $U(\mathfrak{g})$-module Q_χ is called the *generalized Gelfand–Graev module* associated with f.

Since the adjoint action of $\mathfrak{m}$ on $\mathfrak{g}$ extends to an action on $U(\mathfrak{g})$, and since the ideal I_χ is $\mathfrak{m}$-stable, the quotient Q_χ carries a natural $\mathfrak{m}$-module structure. We define

$$Q_\chi^{\mathfrak{m}} := \{\bar{u} \in Q_\chi \mid [x, u] \in I_\chi \text{ for all } x \in \mathfrak{m}\},$$

where $\bar{u}$ denotes the image of $u \in U(\mathfrak{g})$ in Q_χ.

Definition 4.10. The finite W-algebra associated with f is defined by

$$U(\mathfrak{g}, f) := Q_\chi^{\mathfrak{m}}.$$

Equivalently,

$$U(\mathfrak{g}, f) = \{\bar{u} \in Q_\chi \mid [x, u] \in I_\chi \text{ for all } x \in \mathfrak{m}\}.$$

This realization of $U(\mathfrak{g}, f)$ is called the *Whittaker model realization*.

Arakawa proved the following theorem in [Ara07a]:

Theorem 4.11. *There is a natural algebra isomorphism*

$$U(\mathfrak{g}, f) \cong \mathrm{End}_{U(\mathfrak{g})}(Q_\chi)^{\mathrm{op}} \cong Q_\chi^{\mathfrak{m}}.$$

Here op *denotes the opposite algebra structure on* $\mathrm{End}_{U(\mathfrak{g})}(Q_\chi)$*, that is, the multiplication is given by reverse composition:*

$$\varphi \cdot \psi := \psi \circ \varphi.$$

Thus we have given two very different realizations of W-algebras: one geometric and one algebraic. However, by the theorem above, we know that they are the same.

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